Controlling Shareholders and Supermajority Rules*

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Abstract

Conflicted transactions are a first-order governance problem in controlled firms as controllers can extract private benefits unless minorities are protected. Many jurisdictions therefore require approval by disinterested shareholders (majority-of-the-minority, MoM) or heightened supermajority thresholds as safeguards. I develop a signaling model in which a controller privately observes project quality and can distort disclosures at a cost; voting rules set a belief cut-off and thereby govern pooling versus separation. In a baseline "talk-up" environment with standard priors, pooling is smallest under a lenient supermajority, larger under MoM, and largest under a strict supermajority. When beliefs are sluggish—due to noisy signaling, rational inattention, or non-participation—the ordering reverses: pooling is largest under a lenient supermajority, smaller under MoM, and smallest under a strict supermajority. A bar–boundary trade-off drives the ranking: the bar's mass effect versus the boundary's behavioral response.

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1 Introduction

The rise of controlling shareholders represents one of the most significant developments in corporate governance over the past two decades. Even in traditionally diffuse-ownership jurisdictions like the United States and United Kingdom, controlling shareholders have expanded their influence, particularly in very large firms that were once bastions of dispersed ownership (Aminadav and Papaioannou, 2020; Holderness, 2009; Pargendler, 2019; Tallarita, 2024).[1] This shift has sparked intense debate among legal scholars and practitioners, especially following Delaware's 2025 SB 21 reforms, which fundamentally altered the landscape of controlling shareholder transactions. By eliminating the dual requirements for approval by both an independent special committee and a majority of minority shareholders established in *Kahn v. M&F Worldwide Corp.*, 88 A.3d 635 (Del. 2014), and providing de facto immunity to blockholders below 33% of voting power, these reforms have reignited fundamental questions about minority shareholder protection (Bainbridge, 2025; Whitehead, 2025; Khoo and Tallarita, 2025). The implications extend beyond the U.S., as other jurisdictions grapple with similar tensions between facilitating efficient transactions and protecting minority shareholders (Ventoruzzo, 2015; Enriques, 2014; Lim, 2018).

The dominance of controlling shareholder structures is not merely a U.S. phenomenon but reflects global corporate reality. Recent OECD data reveals that concentrated ownership remains the prevailing governance structure across most economies, with controlling shareholders present in the majority of listed companies worldwide (OECD, 2023; De La Cruz et al., 2019). This pattern holds across diverse legal and economic systems: La Porta et al. (1999) document the prevalence of controlling shareholders across 27 wealthy economies; Claessens et al. (2000) find concentrated ownership in over two-thirds of East Asian corporations; while Faccio and Lang (2002) demonstrate that family control dominates Western European firms outside the UK and Ireland.

Despite the economic significance of the agency costs between controlling and minority shareholders—what scholars have termed "Principal-Principal" conflicts—corporate law theory remains disproportionately focused on the board-shareholder agency problem (Goshen and Squire, 2017; Kraakman and Armour, 2017; Anabtawi, 2005). The canonical works in corporate governance—from Bebchuk (2005)'s shareholder empowerment thesis, Blair and Stout (1999)'s team production model, and Bainbridge (2003)'s director primacy theory—primarily address conflicts between professional managers and dispersed shareholders. This theoretical emphasis persists despite empirical evidence that controlling shareholder agency costs may dwarf traditional managerial agency costs in most of the world's corporations (Johnson et al., 2000; Djankov et al., 2008; Atanasov et al., 2014).

When scholars do examine controlling shareholder agency costs, the theoretical framework has largely centered on mechanisms to prevent extraction of private benefits of control (Gilson, 2006; Gilson and Gordon, 2003; Goshen, 2003; Gilson and Schwartz, 2015). The empirical literature documents substantial private benefit extraction: Dyck and Zingales (2004) demonstrate that control premiums average 14% globally but vary dramatically across countries, from -4% in Japan

[1] For instance, Holderness (2009) shows that concentrated ownership is more common than typically assumed in the United States, particularly among smaller public companies.

to 65% in Brazil; Barclay and Holderness (1989) find significant block premiums in U.S. negotiated trades; while Albuquerque and Schroth (2008) show how legal institutions affect the magnitude of private benefits. Yet the effectiveness of legal constraints remains contested. As Choi (2018) argues, controlling shareholders' long-term horizons and reputational concerns may naturally align their interests with minority shareholders (see also Anderson and Reeb, 2003; Villalonga and Amit, 2006). Moreover, controlling shareholders can provide valuable monitoring that reduces managerial agency costs (Shleifer and Vishny, 1986; Burkart et al., 2003), making the necessity and design of legal interventions fundamentally an empirical question.

This paper provides a novel angle to this debate by examining the informational effects of voting mechanisms used to govern conflicted *controller* transactions. Beyond ex-post litigation—whose effectiveness remains debated (Thompson and Thomas, 2016; Choi and Pritchard, 2009)—corporate law relies heavily on ex-ante voting requirements for fundamental corporate changes: mergers and acquisitions (Restrepo, 2021), substantial asset sales , capital structure alterations (Gompers et al., 2010), and most crucially, conflicted transactions involving controlling shareholders (Kraakman and Armour, 2017). The centrality of conflicted transactions is evident in recent Delaware jurisprudence, for instance, in *In re Match Group, Inc. Derivative Litigation*, 2024 WL 1146412 (Del. Ch. Mar. 13, 2024) and *Tornetta v. Musk*, 310 A.3d 430 (Del. Ch. 2024). These transactions represent a first-order concern precisely because they create the starkest conflicts between controlling and minority shareholders (Restrepo, 2021; Djankov et al., 2008).

The standard mechanism for "cleansing" conflicted transactions requires approval by non-conflicted shareholders, effectively excluding the controlling shareholder from voting (Subramanian, 2005). This "majority of the minority" (MoM) rule has become a cornerstone of minority protection across jurisdictions (Goshen and Hamdani, 2019). However, corporate law employs various other voting thresholds that functionally resemble MoM but operate through different mechanics (Easterbrook and Fischel, 1983). Special resolutions in the United Kingdom (UK), requiring 75% approval, create varying effective thresholds depending on the controller's stake—if a controlling shareholder holds 60% of shares (assuming one-share-one-vote), minorities need only garner 15% support, making the requirement less stringent than MoM.[2] Conversely, supermajority rules requiring extreme consensus, such as the 90% threshold for compulsory share acquisitions in squeeze-out transactions for the UK and Hong Kong, can demand greater minority support than simple MoM.[3] These variations in voting rules may have profound implications for corporate governance outcomes; however, it is not obvious whether a MoM regime has superior efficiency properties relative to supermajority regimes.

[2]See Section 283 of the UK Companies Act. Australia similarly requires "Special Resolutions" with 75% of eligible votes for actions such as modifying the company Constitution or selective share reductions (See Section 9 read with Section 250MA of the Australian Companies Act).

[3]See Section 979 of the UK companies Act and Section 694 of the Hong Kong Companies Ordinance.

2 Literature Review

Prior theoretical research has often compared Majority-of-the-Minority (MoM) and supermajority rules through the lens of Type I and Type II errors—approving bad transactions versus blocking good ones (Thompson, 1995)—typically in environments where information asymmetry plays a secondary role (Maug and Rydqvist, 2009). Other strands of the voting literature, from Harris and Raviv (1988) on one-share-one-vote to Grossman and Hart (1988) on security voting, largely abstract from strategic communication between informed controllers and uninformed minorities. By contrast, this paper places *information transmission* at center stage: voting thresholds discipline a privately informed controller's disclosure policy, and the communication technology (e.g., costly signaling) matters for how thresholds map into equilibrium outcomes (Crawford and Sobel, 1982; Farrell and Rabin, 1996; Spence, 1973; Milgrom and Roberts, 1986).

Our contribution is to develop a model of adverse selection with costly signaling in which voting thresholds endogenously determine whether persuasion is *up-talk* or *down-talk*, and thereby shape the location and size of pooling regions. The analysis provides a unifying "bar–boundary" framework: a change in the voting bar mechanically sweeps probability mass into the proposal set while simultaneously triggering a behavioral response of the marginal type (Milgrom and Shannon, 1994; Kartik, 2009). This trade-off delivers a rule ranking that depends on the information environment: under identity beliefs and standard priors (e.g., uniform/log–concave), pooling is smallest under a lenient supermajority, larger under MoM, and largest under a strict supermajority; with sluggish beliefs—motivated by noisy signaling, rational inattention, or endogenous non-participation—the ordering reverses (Blume et al., 2007; Sims, 2003; Matějka and Tabellini, 2021; Palfrey and Rosenthal, 1983; Osborne et al., 2000; Levine and Palfrey, 2007). In a diversion-dominant regime in which private benefits exceed minorities' marginal returns, multiple equilibria can arise, rendering the ranking ambiguous. Across these cases, the paper's contribution is to show how voting thresholds govern equilibrium communication—rather than merely error probabilities—and to characterize when comparative statics flip as beliefs become less responsive.

Our approach complements work on strategic voting and information aggregation (Levit et al., 2023, 2024; Meirowitz and Pi, 2022) by emphasizing the controller's disclosure incentives toward uninformed minorities, and relates to similar research on voting mechanisms and corporate voting rules (Gershkov et al., 2017; Easterbrook and Fischel, 1983; Miller, 2021; Harris and Raviv, 2010) by highlighting the informational (rather than purely distributive) consequences of thresholds. Whereas papers such as Meirowitz and Pi (2023) study endogenous information acquisition, the present framework focuses attention on how thresholds reshape disclosure costs and, in turn, the equilibrium alignment of pooling versus separation.

Empirically (and anecdotally), the evidence on MoM and supermajority rules is mixed: case analyses report MoM approvals that "did not hurt, but did not do much good either" (Rock, 2018), while Israeli data show institutional investors often support insider-sponsored transactions even when minority voting power is formally enhanced, yielding ambiguous real effects (Hamdani and Yafeh, 2013). Critics point to frictions—conflicts of interest, information asymmetry, and

empty voting—and propose alternatives such as "Majority of the Quality Shareholders" (Cunningham, 2021; Hu and Hamermesh, 2025; Bebchuk and Kastiel, 2024). Other work describes contexts in which MoM can facilitate negotiated outcomes acceptable to minorities (Booth, 2013), and evidence from China links MoM disapprovals of related-party transactions to efficiency improvements, alongside concerns about "investment myopia" around votes (Cao et al., 2024; Lin et al., 2025). The model here rationalizes such heterogeneity: differences in priors, belief responsiveness, and private-benefit regimes can reverse the ordering of pooling and separation across rules, implying that the comparative performance of MoM and supermajority thresholds is context-dependent (Beim et al., 2016).

The remainder of the paper proceeds as follows. Section 3 presents the model setup, detailing the information structure and cost assumptions. Section 4 characterizes equilibrium communication under different voting rules. Sections 5 and 6 analyze informational efficiency and welfare across thresholds, including the comparative-statics reversals under sluggish beliefs and the multiplicity in diversion-dominant environments. Section 7 concludes.

3 The Model: Primitives and Assumptions

In this section, I characterize the key model that motivates the rest of my analysis. Subsequent theorems or proof will cite only the numbered statements below.

3.1 Controlling Shareholders

3.1.1 Project Quality

There exists a single *controlling shareholder* (CS) who permanently owns a strictly positive but non-unanimous voting stake $\alpha \in (0, 1)$, while a large but finite number N of symmetric minority shareholders–which will be discussed later–hold the residual $1 - \alpha$.

The quality of a project is denoted by the variable θ. It is realised once, observed privately by the CS, and then remains fixed throughout the game. Importantly, θ is distributed $\theta \sim F$, and F is continuous on the compact interval $\left[\underline{\theta}_0, \overline{\theta}_0\right] \subset \mathbb{R}$, where the support endpoints satisfy

$$[\,\underline{\theta}_0,\ \overline{\theta}_0], \qquad \underline{\theta}_0 < 0 < \overline{\theta}_0. \tag{P1}$$

We assume the following:

Assumption 1 (Log–concavity and monotone shoulders (P2))**.** *The type distribution F has a density $f := F'$ that is strictly positive on $[\underline{\theta}_0, \overline{\theta}_0]$ and* log–concave*:*

$$f(x) > 0 \quad \textit{and} \quad \log f(\cdot) \textit{ is concave on } [\underline{\theta}_0, \overline{\theta}_0].$$

Let $u_{\mathrm{MoM}} \in (\underline{\theta}_0, \overline{\theta}_0)$ denote the (unique) mode of f. Then f has monotone shoulders*: it is* nondecreasing *on $[\underline{\theta}_0,\ u_{\mathrm{MoM}}]$ (left shoulder) and* nonincreasing *on $[\ u_{\mathrm{MoM}},\ \overline{\theta}_0]$ (right shoulder), where the belief-cutoff u_{MoM} is defined later in V3 (Section 3.3.3).*

3.1.2 Private Benefits

Implementing the project gives the controller transferable private benefit $B(\theta)$ on top of the pro-rata payoff $\alpha\theta$. We assume

$$B \ : \ [\,\underline{\theta}_0, \overline{\theta}_0] \ \longrightarrow \ \mathbb{R}, \qquad B \in C^1, \qquad 0 < B'(\theta) < 1 - \alpha \quad \big(\forall\, \theta\big). \tag{P3}$$

where the level can be normalized by setting $B(\underline{\theta}_0) = 0$.

Because the marginal diversion is always strictly below the minorities' residual payoff share due to assumption P3, the net minority payoff

$$(1-\alpha)\,\theta \ - \ B(\theta)$$

is strictly increasing over the entire support $[\underline{\theta}_0, \overline{\theta}_0]$, a property used in many of the single-crossing equilibrium-selection arguments that follow (Milgrom and Shannon, 1994).

If $B'(\theta) > 1 - \alpha$, the acceptance criteria for minority shareholders reverses, so single-crossing arguments are no longer guaranteed. We address these complications in Section 6.1.

3.1.3 Signalling technology

After observing θ, the controller sends a publicly observable message $m \in \mathbb{R}$. Signalling incurs an out-of-pocket cost $C(m, \theta)$ that depends only on the *gap* $\Delta \ := m - \theta$. We adopt a smooth, convex, gap–based specification with a linear "lip" and a quadratic component, where we define the parameters $\eta > 0$ and $\kappa \geq 0$ such that, for all $(m, \theta) \in \mathbb{R}^2$:

$$C(m,\theta) \ = \ \eta\,|m-\theta| \ + \ \frac{\kappa}{2}\,(m-\theta)^2, \qquad \text{with the normalization } C(\theta,\theta) = 0. \tag{S1}$$

Equivalently, writing $\Delta \ := m - \theta$,

$$C(m,\theta) = \begin{cases} \eta\,(\theta - m) \ + \ \dfrac{\kappa}{2}\,(m-\theta)^2, & m < \theta, \\ 0, & m = \theta, \\ \eta\,(m - \theta) \ + \ \dfrac{\kappa}{2}\,(m-\theta)^2, & m > \theta. \end{cases} \tag{S2}$$

S3: Convexity and symmetry: C is convex in the gap Δ (sum of the convex map $\Delta \mapsto |\Delta|$ and the quadratic $\Delta \mapsto \frac{\kappa}{2}\Delta^2$) and *symmetric* in the sense $C(\theta + \Delta, \theta) = C(\theta - \Delta, \theta)$.

S4: Differentiability and marginal costs: C is C^∞ on each wedge $\{\Delta \neq 0\}$ and has a (single) kink at $\Delta = 0$. The partial derivatives with respect to the message and the type exist for $\Delta \neq 0$ and satisfy

$$C_m(m,\theta) = \begin{cases} -\eta + \kappa\Delta, & \Delta < 0, \\ \eta + \kappa\Delta, & \Delta > 0, \end{cases} \qquad C_\theta(m,\theta) = -\,C_m(m,\theta) = \begin{cases} \eta - \kappa\Delta, & \Delta < 0, \\ -\eta - \kappa\Delta, & \Delta > 0, \end{cases} \tag{S4}$$

with subgradient $C_m(0) \in [-\eta, \eta]$ at $\Delta = 0$. In particular,

$$C_m(0^-) = -\eta, \qquad C_m(0^+) = +\eta,$$

so both the up– and down–wedge display a strictly positive *linear lip* of size η at the truth.

S5: Monotonicity on (convex) wedges: On the up–wedge ($\Delta > 0$), the marginal signaling cost is strictly positive and strictly increasing in the size of the lie:

$$C_m(\Delta) = \eta + \kappa\Delta \quad \text{with} \quad \frac{d}{d\Delta} C_m(\Delta) = \kappa \geq 0.$$

On the down–wedge ($\Delta < 0$), the marginal cost is strictly negative and strictly decreasing as the lie moves further downward:

$$C_m(\Delta) = -\eta + \kappa\Delta \leq -\eta < 0, \quad \frac{d}{d\Delta} C_m(\Delta) = \kappa \geq 0.$$

Hence $|C_m(\Delta)| \geq \eta$ for all $\Delta \neq 0$, and $|C_m(\Delta)|$ is (weakly) increasing in $|\Delta|$.

S6 (Tail–growth, with bounded–gap caveat). The signaling cost depends on the gap $\Delta := m - \theta$ via a convex, wedge–symmetric map $C(\Delta)$ with $C(0) = 0$ and C strictly increasing in $|\Delta|$.

(a) Unbounded–gap case. If for every θ the feasible message set allows a sequence $\{m_n\}$ with $|m_n - \theta| \to \infty$, then

$$C(\Delta) \xrightarrow[|\Delta|\to\infty]{} \infty,$$

so sufficiently large lies are eventually prohibitive.

(b) Bounded–gap case (when tail–growth may not bite). If the feasible set and posteriors imply a *gap envelope*

$$\bar{\Delta}(u) \;=\; m^{\text{req}}(u) - u \quad \text{is bounded on } u \in (\underline{\theta}_0, \overline{\theta}_0),$$

then $|\Delta|$ cannot grow arbitrarily and the "tail–growth" force in (a) need not trigger separation. In this case, separation requires a finite-bar dominance test,

$$\Xi_h(u) \;=\; \alpha u + B(u) - C\big(\bar{\Delta}(u)\big) \;\leq\; 0 \quad \text{for some interior } u,$$

rather than appeal to $|\Delta| \to \infty$.

S7: Type derivative: Because C depends on (m, θ) only through $\Delta = m - \theta$, we have $C_\theta = -\,C_m$ for $\Delta \neq 0$. No sign restriction on C_θ is imposed, and all comparative–statics that follow refer directly

to C_m on the relevant wedge.

3.1.4 Controlling Shareholder's Utility Function

Given the primitives in Sections 3.1.1 and 3.1.3, and the voting cut-off in (V2), define the rule-specific approval indicator

$$\mathbf{1}\{\text{approve}(m)\} \;:=\; \mathbf{1}\Big\{(1-\alpha)\,\mu(m) \;-\; \beta(m) \;\geq\; \kappa^{\text{rule}}(\lambda)\Big\} \;=\; \mathbf{1}\{\mu(m) \;\geq\; u\},$$

where u is the belief cut-off from (V3). The controlling shareholder chooses a message anticipating minority voting:

$$\max_{m}\; U_{CS}(m,\theta) \;=\; \mathbf{1}\{\text{approve}(m)\}\cdot\Big(\alpha\,\theta + B(\theta)\Big) \;-\; C(m,\theta). \tag{C1}$$

3.1.5 Controlling Shareholder's Participation Constraint

Absent additional structure, Assumption P3 together with the payoff criterion in (C1) implies that a controller has little incentive to initiate projects with $\theta \leq 0$. To ensure well–posed participation at the voting cut–off, we adopt the affine specification $B(\theta) = b_0 + b_1\theta$ with $b_1 \in [0,\, 1-\alpha)$ and choose b_0 large enough that the controller finds it (weakly) profitable to proceed at the bar.[4] This is encoded in the following assumption.

Assumption 2 (Controller participation at the rule cut-off (CPC) and interior scope)**.** *Maintain Assumptions P1–P3, S1–S4, and M1. In the baseline, take $B'(\theta) \in (0,\, 1-\alpha)$ and $B(\theta) = b_0 + b_1\theta$ with $b_1 \in [0,\, 1-\alpha)$. For any voting environment that induces a belief cut-off u via (V2)–(V3), require the controller's participation constraint at the cut-off u to hold:*

$$\alpha\, u + B(u) \;\geq\; 0. \tag{CPC}$$

Under $B(\theta) = b_0 + b_1\theta$ this is equivalent to

$$(\alpha + b_1)\, u + b_0 \;\geq\; 0 \qquad\Longleftrightarrow\qquad u \;\geq\; u_{\text{CPC}} \;:=\; -\,\frac{b_0}{\alpha + b_1}.$$

Because $\alpha + b_1 > 0$, (CPC) imposes a lower bound u_{CPC} on admissible cut-offs. In the baseline, b_0 is chosen so that $u \geq u_{\text{CPC}}$ along every interior corridor of cut-offs considered, ensuring the controller participates at the bar.

[4] Henceforth, we refer to the "bar" as the belief cut-off, u defined in V3.

3.2 Minority Shareholders

3.2.1 Minority Shareholder Voting and Posterior Beliefs

As discussed in Section 3.1.1, a large but finite number N of symmetric minority shareholders collectively hold the residual $1-\alpha$ of the project's valuation, θ. To simplify notation, I define the variables s and λ:

$$s \;=\; \frac{1-\alpha}{N}, \qquad \lambda \;:=\; \frac{s}{1-\alpha} \;=\; \frac{1}{N} \;\in\; (0,1).$$

Hence each minority shareholder owns the fraction λ of the residual stake and, correspondingly, is exposed to a λ-proportion of any private benefits extracted by the controlling shareholder. We assume $\sum_{n=1}^{N} \lambda\, B(\theta) = B(\theta)$, implying that the controlling shareholder's extraction of private benefits is zero-sum. Except where noted, we will henceforth treat the case of a single representative minority shareholder, without loss of generality.

3.2.2 Beliefs and Updating

For each minority shareholder, a *belief system* specifies, for each publicly observed message $m \in \mathbb{R}$, a posterior distribution $\rho(\cdot \mid m) \in \Delta([\underline{\theta}_0, \overline{\theta}_0])$ over types. On the equilibrium path, beliefs are obtained by Bayes' rule from the prior F and the sender's strategy σ;[5] off the equilibrium path we adopt the Grossman–Perry *credible-beliefs* refinement (Appendix A.2): upon observing an unexpected message, minorities place probability 1 on the set of types that (weakly) benefit most from that deviation (Grossman and Perry, 1986).

We summarize posteriors by their expectations

$$\mu(m) \;:=\; \mathbb{E}[\theta \mid m], \qquad \beta(m) \;:=\; \mathbb{E}[B(\theta) \mid m],$$

which by assumptions P1 and P3 always lie in $[\underline{\theta}_0, \overline{\theta}_0]$ and $[B(\underline{\theta}_0), B(\overline{\theta}_0)]$, respectively. In other words, $\mu(m)$ reflect's the minority shareholder's conditional expectation of the project's valuation when they observe the message m. We assume that:

$\mu(m)$ is continuous, strictly increasing, and invertible; accordingly, μ^{-1} exists and is continuous. (M1)

We further impose the following tail–limit regularity on beliefs: as messages become arbitrarily small or large, the posterior mean converges to the endpoints of the valuation support, i.e.,

$$\lim_{m\to-\infty} \mu(m) = \underline{\theta}_0, \qquad \lim_{m\to+\infty} \mu(m) = \overline{\theta}_0. \tag{M1$'$}$$

[5] See Section 3.3.5 for strategy notation.

3.2.3 Expected Utility from Approving (Interim Stage)

Following (Levit et al., 2024), we endow each minority investor i with a privately known bias b_i that perturbs her cut-off for approving the proposal. The bias is normalised as

$$b_i \sim \text{Uniform}[-1,\ 1], \qquad \mathbb{E}[b_i] = 0, \tag{M2}$$

independent of θ and distributed across investors. Using the posterior summaries from Section 3.2.2, investor i's interim utility upon approving is

$$\mathbb{E}[u_i \mid m] = \lambda\Big[(1-\alpha)\,\mu(m) - \beta(m)\Big] + b_i, \tag{M3}$$

with $G(b) = (b+1)/2$ for $b \sim U[-1,1]$.

3.2.4 Yes/No rule and Pivotal bias

We define

$$t_\lambda(m) := -\lambda\Big[(1-\alpha)\mu(m) - \beta(m)\Big].$$

Thus, a minority shareholder i votes "Yes" if and only if $b_i \geq t_\lambda(m)$. Accordingly, the fraction of "Yes" votes is

$$1 - G\big(t_\lambda(m)\big) = \frac{1 - t_\lambda(m)}{2}, \tag{M4}$$

given the distribution of $b_i \sim U[-1,1]$, for $t_\lambda(m) \in [-1,1]$ ($t_\lambda(m)$ saturates at 0 or 1 outside).

3.3 Voting Rules

3.3.1 Definition

We consider two types of Voting Rules.

1. **Majority of the Minority (MoM).** A proposal is implemented if and only if strictly more than 0.5 of the $(1-\alpha)$ minority shares vote "Yes."

2. **Super-majority (SM).** Fix a quota $S \in (0.5, 1)$. Implementation requires at least a fraction S of *all* shares in favour. Equivalently, the fraction of minority votes needed is

$$\pi \;=\; \frac{S-\alpha}{1-\alpha}. \tag{V1}$$

   An SM rule is defined to be *lenient* when $\pi < 0.5$ and *stringent* when $\pi > 0.5$.

With these primitives fixed, all subsequent analysis derives equilibrium signalling and voting outcomes and compares welfare across the two rule families.

3.3.2 Cut-offs under MoM and super-majority S

For the MoM rule, the cutoff point is defined by the minority shareholder who is indifferent between voting "Yes" and "No" for a given proposal, conditional on the message m sent by the CS. As such, given expressions M4 and V1, under the MoM rule, shareholders vote "Yes" iff:

$$\frac{1 - t_\lambda(m)}{2} \;\geq\; \tfrac{1}{2} \;\Longrightarrow\; t_\lambda(m) \leq 0 \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta(m) \;\geq\; 0.$$

Similarly, under the SM rule, let $\pi = (S-\alpha)/(1-\alpha)$. Requiring $(1 - t_\lambda(m))/2 \geq \pi$ as per expressions M4 and V1 gives the condition where shareholders vote "Yes" iff:

$$t_\lambda(m) \leq 1 - 2\pi \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta(m) \;\geq\; -\frac{1-2\pi}{\lambda}.$$

We can then define the rule-specific cutoff

$$\underline{\kappa}^{(\lambda)}_{\text{rule}} := -\frac{k_{\text{rule}}}{\lambda}, \quad k_{\text{MoM}} = 0, \quad k_{\text{SM}} = 1 - 2\pi,$$

so that in both cases minorities approve iff:

$$(1-\alpha)\,\mu(m) - \beta(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\text{rule}}. \tag{V2}$$

3.3.3 Belief Cut-offs u

Fix a voting rule and exposure λ. Let $\kappa^{(\lambda)}_{\text{rule}}$ be the constant in (V2). Define the minimal persuasive message which would pass the proposal given a voting rule:

$$m_{\text{req}} := \inf\left\{ m \in \mathbb{R} \;:\; (1-\alpha)\mu(m) - \beta(m) \;\geq\; \kappa^{(\lambda)}_{\text{rule}} \right\}, \qquad u := \mu(m_{\text{req}}). \tag{V3}$$

By (M1), μ is strictly increasing and continuous, so m_{req} exists and is unique with $m_{\text{req}} = \mu^{-1}(u)$. Given that $\mu(m) = \mathbb{E}[\theta \mid m]$, u is the conditional expected valuation of the pivotal shareholder who induces the proposal to pass. We refer to u as the belief cut-off/threshold for passage.

3.3.4 Regime Orientation

When private benefits are modest, $0 < B'(\theta) < 1 - \alpha$ (*payoff–dominant*), minorities' net payoff $(1-\alpha)\theta - B(\theta)$ rises with true quality, so they welcome higher perceived value. The approval

test is $\mu(m) \geq u$, any pooling (if present) lies *below* u, and persuasion is *upward* (types just below u talk up to $m_{\text{req}}(u)$). Equivalently, with $\Phi(m) := (1-\alpha)\mu(m) - \beta(m)$, Φ is strictly increasing in μ, yielding a unique $u \in (\underline{\theta}_0, \overline{\theta}_0)$.

When private benefits are steep, $B'(\theta) > 1 - \alpha$ (*diversion–dominant*), Φ is strictly decreasing in μ, the approval test flips to $\mu(m) \leq u$, any pooling lies *above* u, and persuasion is *downward*. In both regimes the minimal persuasive message is $m_{\text{req}} = \mu^{-1}(u)$; only the inequality direction in μ reverses across regimes.

3.3.5 Interior cut-offs and strategy notation

Standing assumption (interior cut-off): Unless stated otherwise, $u \in (\underline{\theta}_0, \overline{\theta}_0)$; the degenerate cases $u \notin (\theta_0, \bar{\theta}_0)$ (all-Yes/all-No) are collected in Corollary 1 (Appendix).

Strategy notation: The controller's message strategy depends on type θ and on the belief environment through u, such that $m = \sigma_u(\theta)$, or $m(\theta; u)$ when the u-dependence matters.

3.3.6 Closed-Form Cut-Offs with Affine Private Benefits

The belief cut-off u depends on the shape of the private-benefit function $B(\theta)$. To obtain closed-form expressions, assume the controller's private benefits are *affine*:

$$B(\theta) \;=\; b_0 + b_1\theta, \qquad b_1 \in (0,\, 1-\alpha)\ (\text{P3}), \quad b_0 \in \mathbb{R}.$$

Given any publicly observed message m, the posterior summaries from Section 3.2.2 satisfy

$$\beta(m) \;=\; \mathrm{E}\big[B(\theta) \mid m\big] \;=\; b_0 + b_1\,\mu(m).$$

Substituting this into the common voting inequality (V2),

$$(1-\alpha)\,\mu(m) \;-\; \beta(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\text{rule}},$$

yields the *linear* acceptance condition

$$(1-\alpha-b_1)\,\mu(m) \;-\; b_0 \;\geq\; \underline{\kappa}^{(\lambda)}_{\text{rule}}.$$

Because $1-\alpha-b_1 > 0$ by P3, this is equivalent to the belief threshold

$$\mu(m) \;\geq\; u_{\text{rule}}(\lambda) \;:=\; \frac{\underline{\kappa}^{(\lambda)}_{\text{rule}} + b_0}{1-\alpha-b_1} \tag{V4}$$

for each voting rule.

Recalling the rule-specific constants after (V2), $\underline{\kappa}^{(\lambda)}_{\text{MoM}} = 0$ and $\underline{\kappa}^{(\lambda)}_{\text{SM}} = -(1-2\pi)/\lambda$, V4 gives

$$u_{\text{MoM}} \;=\; \frac{b_0}{1-\alpha-b_1}, \qquad u_{\text{SM}}(\pi,\lambda) \;=\; \frac{b_0 - \frac{1-2\pi}{\lambda}}{1-\alpha-b_1}.$$

Hence, relative to the $b_0 = 0$ benchmark, a positive intercept $b_0 > 0$ uniformly *raises* both cut-offs by $b_0/(1-\alpha-b_1)$. In particular, the MoM cut-off need not be zero when $b_0 \neq 0$.

Keeping (α, b_1) fixed,

$$\frac{\partial u_{\text{SM}}}{\partial \pi} = \frac{2/\lambda}{1-\alpha-b_1} > 0$$

Thus, stricter quotas (higher π) tighten the belief cut-offs u required to pass a given proposal. In a separating equilibrium (so that $\mu(m) = \theta$), the belief thresholds above translate directly into type thresholds, $\theta \geq u_{\text{rule}}(\lambda)$. In a pooling equilibrium, minorities compare the pooled posterior mean with $u_{\text{rule}}(\lambda)$ from V4.

3.4 Vanishing Exposure as $\lambda \to 0$

Under MoM, the voting inequality is $(1-\alpha)\,\mu(m) - \beta(m) \geq 0$ (cf. (V2)), which does not depend on λ; hence ownership dispersion has no effect on the approval set. Under a lenient SM rule, $\pi < \frac{1}{2}$, the rule constant in (V2) equals $\underline{\kappa}^{(\lambda)}_{\text{SM}} = -(1-2\pi)/\lambda \to -\infty$ as $\lambda \downarrow 0$, so the hurdle becomes vacuous in the atomistic limit. Under a stringent SM rule, $\pi > \frac{1}{2}$, $\underline{\kappa}^{(\lambda)}_{\text{SM}} = -(1-2\pi)/\lambda = (2\pi-1)/\lambda \to +\infty$ as $\lambda \downarrow 0$, so no posterior can clear the hurdle. The proposition formalizes these limits.

Proposition 1 (Limit approval outcomes as $\lambda \to 0$)**.** *Fix a voting rule and work under* (P1)–(P3) *and* (M1). *Let* $\mu(m) = \mathrm{E}[\theta \mid m] \in [\underline{\theta}_0, \overline{\theta}_0]$ *and* $\beta(m) = \mathrm{E}[B(\theta) \mid m] \in [B(\underline{\theta}_0),\, B(\overline{\theta}_0)]$. *Then:*

- *MoM* ($\pi = \frac{1}{2}$)*. The approval inequality is independent of* λ*; the approval set (hence the implementation allocation for any equilibrium consistent with* (V2)*) is unchanged as* $\lambda \to 0$.
- *Lenient SM* ($\pi < \frac{1}{2}$)*. There exists* $\bar{\lambda} > 0$ *such that for all* $\lambda < \bar{\lambda}$, (V2) *holds for every message* m *and approval is certain:* $\Pr(\textit{approval} \mid \lambda < \bar{\lambda}) = 1$.
- *Stringent SM* ($\pi > \frac{1}{2}$)*. There exists* $\bar{\lambda} > 0$ *such that for all* $\lambda < \bar{\lambda}$, (V2) *fails for every message* m *and rejection is certain:* $\Pr(\textit{approval} \mid \lambda < \bar{\lambda}) = 0$.

Proof. By (P1) and the definition of posteriors, for every m,

$$\mu(m) \in [\underline{\theta}_0, \overline{\theta}_0], \qquad \beta(m) \in [B(\underline{\theta}_0),\, B(\overline{\theta}_0)].$$

Hence the left–hand side of the voting inequality (V2) is uniformly bounded:

$$(1-\alpha)\,\underline{\theta}_0 - B(\overline{\theta}_0) \;\leq\; (1-\alpha)\,\mu(m) - \beta(m) \;\leq\; (1-\alpha)\,\overline{\theta}_0 - B(\underline{\theta}_0). \tag{V4'}$$

MoM. For MoM, $\underline{\kappa}^{(\lambda)}_{\text{MoM}} = 0$ (see (V2)), which is independent of λ. Thus, for any λ, the set $\{m : (1-\alpha)\mu(m) - \beta(m) \geq 0\}$ is the same, so approval decisions are unaffected as $\lambda \to 0$.

Lenient SM. For $\pi < \frac{1}{2}$, $\underline{\kappa}_{\mathrm{SM}}^{(\lambda)} = -(1-2\pi)/\lambda \to -\infty$ as $\lambda \downarrow 0$. Choose $\bar{\lambda} > 0$ such that

$$-\frac{1-2\pi}{\bar{\lambda}} \;\leq\; (1-\alpha)\,\underline{\theta}_0 - B(\overline{\theta}_0).$$

Then for any $\lambda < \bar{\lambda}$, $\underline{\kappa}_{\mathrm{SM}}^{(\lambda)} \leq -(1-2\pi)/\bar{\lambda} \leq (1-\alpha)\,\underline{\theta}_0 - B(\overline{\theta}_0)$. By (V4′), $(1-\alpha)\mu(m) - \beta(m) \geq \underline{\kappa}_{\mathrm{SM}}^{(\lambda)}$ for every m, so (V2) holds for all messages and approval is certain.

Stringent SM. For $\pi > \frac{1}{2}$, $\underline{\kappa}_{\mathrm{SM}}^{(\lambda)} = -(1-2\pi)/\lambda = (2\pi-1)/\lambda \to +\infty$ as $\lambda \downarrow 0$. Pick $\bar{\lambda} > 0$ such that

$$\frac{2\pi-1}{\bar{\lambda}} \;\geq\; (1-\alpha)\,\overline{\theta}_0 - B(\underline{\theta}_0).$$

Then for any $\lambda < \bar{\lambda}$, $\underline{\kappa}_{\mathrm{SM}}^{(\lambda)} \geq (2\pi-1)/\bar{\lambda} \geq (1-\alpha)\,\overline{\theta}_0 - B(\underline{\theta}_0)$. By (V4′), $(1-\alpha)\mu(m) - \beta(m) \leq \underline{\kappa}_{\mathrm{SM}}^{(\lambda)}$ for every m, so (V2) fails for all messages and rejection is certain.

This establishes all three claims. □

As ownership becomes extremely dispersed ($\lambda \to 0$), each minority investor's personal exposure to the project's payoff and to any diversion becomes negligible, so her private benefit from screening is too small to justify costly opposition—the classic free-rider logic in corporate control (Shleifer and Vishny (1986)). Under a lenient super-majority ($\pi < \frac{1}{2}$), the effective voting hurdle on minority support collapses in the atomistic limit, so even weak, lightly informed "Yes" support suffices and all proposals pass. Under a stringent super-majority ($\pi > \frac{1}{2}$), the hurdle explodes, and with atomistic owners unwilling to incur screening costs or coordinate, no proposal can clear the bar. By contrast, Majority-of-the-Minority fixes the belief cut-off independently of λ, placing it exactly at the knife-edge: greater dispersion neither weakens nor tightens the approval condition, so the approval set is unchanged as $\lambda \to 0$. However, as shown below, this knife-edge neutrality does *not* carry over to the ordering of pooling regions: even holding dispersion fixed, different rules can generate systematically larger or smaller pooling masses once beliefs and signaling costs are taken into account.

4 The Model: Signalling Incentives and Equilibrium Structure

We begin by establishing four general lemmas that apply to *any* voting rule in the baseline "talk–up" regime.

4.1 Lying Profit Function

We reserve $u \in (\underline{\theta}_0, \overline{\theta}_0)$ for the rule-specific belief cut-off from (V3) and θ for types. First, we redefine the "minimal persuasive message" in V3 as the "cheapest effective lie" $m_{\text{req}}(u)$–the lowest-cost message required to induce approvals by the pivotal minority shareholders given their belief thresholds.

Second, define the CS's net gain from lying as a "lying-profit function", $\Lambda(\theta; u)$, which is a function both of the project's valuation θ and the cutoff u. A "lying-profit function" determines the payoff to the CS from misrepresenting the true valuation of the project, θ, such that minority shareholders will accept the proposal in equilibrium. We define $\Lambda(\theta; u)$ as:

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big).$$

When $\Lambda(\theta; u) > 0$ for types just below u, at least some such types strictly prefer to shade up to $m_{\text{req}}(u)$ to secure approval.

4.2 General lemmas

Lemma 1 (Sensitivity of Λ to the rule cut-off)**.** *Fix an interior belief cut-off $u \in (\underline{\theta}, \bar{\theta})$ and let $m^{\text{req}}(u) := \mu^{-1}(u)$. For every (θ, u) with $m^{\text{req}}(u) \neq \theta$,*

$$\frac{\partial \Lambda}{\partial u}(\theta; u) \;=\; -\,C_m\big(m^{\text{req}}(u), \theta\big)\,\mu^{-1\prime}(u).$$

Because $\mu^{-1\prime}(u) > 0$ by (M1)*, the sign is*

$$\operatorname{sign}\Big(\tfrac{\partial \Lambda}{\partial u}\Big) \;=\; -\operatorname{sign}\big(C_m(m^{\text{req}}(u), \theta)\big) \;=\; \operatorname{sign}\big(\theta - m^{\text{req}}(u)\big).$$

In particular, $\frac{\partial \Lambda}{\partial u} < 0$ whenever $m^{\text{req}}(u) > \theta$ and $\frac{\partial \Lambda}{\partial u} > 0$ whenever $m^{\text{req}}(u) < \theta$. At the boundary $m^{\text{req}}(u) = \theta$, the one-sided derivatives in u exist and have the corresponding signs.

Proof. By definition $m^{\text{req}}(u) = \mu^{-1}(u)$, so $\frac{\partial m^{\text{req}}}{\partial u} = \mu^{-1\prime}(u) > 0$ by (M1). Using the definition $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(m^{\text{req}}(u), \theta)$ and applying the chain rule to the composition $u \mapsto C(m^{\text{req}}(u), \theta)$ (which is C^1 off the diagonal by (S1) when $m^{\text{req}}(u) \neq \theta$) yields

$$\frac{\partial \Lambda}{\partial u}(\theta; u) = -\,C_m\big(m^{\text{req}}(u), \theta\big)\,\frac{\partial m^{\text{req}}}{\partial u} \;=\; -\,C_m\big(m^{\text{req}}(u), \theta\big)\,\mu^{-1\prime}(u).$$

Hence $\operatorname{sign}(\partial\Lambda/\partial u) = -\operatorname{sign}\big(C_m(m^{\text{req}}(u), \theta)\big)$. By (S4) and (S2), on the down–wedge ($m < \theta$) we have $C_m(m, \theta) = -\eta + \kappa(m - \theta) < 0$, while on the up–wedge ($m > \theta$) we have $C_m(m, \theta) = +\eta + \kappa(m - \theta) > 0$; the two meet at $m = \theta$ with the one–sided limits $\mp\eta$. Substituting $m = m^{\text{req}}(u)$

gives the stated regional sign: negative when $m^{\text{req}}(u) > \theta$ and positive when $m^{\text{req}}(u) < \theta$.[6] $\square$

Lemma 2 (Single–crossing of the lying–profit function)**.** *Fix any interior belief cut–off $u \in (\underline{\theta}_0, \overline{\theta}_0]$ and define*

$$m^{\text{req}}(u) := \mu^{-1}(u), \qquad \Lambda(\theta; u) := \alpha\,\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Under (P3) *and* (S4) *(with $\alpha > 0$), the map $\theta \mapsto \Lambda(\theta; u)$ is strictly increasing on $[\underline{\theta}_0, u]$. Consequently, for each fixed u the equation $\Lambda(\theta; u) = 0$ has* at most one *solution $\theta^*(u) \in [\underline{\theta}_0, u]$. If in addition there exist θ_1, θ_2 with $\Lambda(\theta_1; u) < 0 < \Lambda(\theta_2; u)$, then there is a* unique *$\theta^*(u) \in (\theta_1, \theta_2)$ with $\Lambda(\theta^*(u); u) = 0$.*

Proof. By (S1) and (M1), C is C^1 off the diagonal and μ^{-1} is continuous, so Λ is continuously differentiable in θ on $[\underline{\theta}_0, u]$, with

$$\Lambda_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta\big(m^{\text{req}}(u), \theta\big).$$

On the relevant region $\theta \in [\underline{\theta}_0, u]$ we have $m_{\text{req}}(u) \geq \theta$ (up–wedge), hence by (S2)–(S4)

$$C_\theta\big(m_{\text{req}}(u), \theta\big) = -(\eta + \kappa\Delta) \leq -\eta < 0 \quad \text{for } \Delta := m_{\text{req}}(u) - \theta \geq 0.$$

Therefore

$$\Lambda_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta\big(m_{\text{req}}(u), \theta\big) \;\geq\; \alpha + B'(\theta) + \eta \;>\; 0,$$

using $B'(\theta) > 0$ from (P3). Thus $\theta \mapsto \Lambda(\theta; u)$ is strictly increasing on $[\underline{\theta}_0, u]$. If $\Lambda(\theta_1; u) < 0 < \Lambda(\theta_2; u)$ for some $\theta_1 < \theta_2$, continuity of $\Lambda(\cdot; u)$ in θ yields existence of a zero in (θ_1, θ_2), and strict monotonicity yields uniqueness. $\square$

Lemma 3 (Monotonicity of the zero-crossing in the cut–off)**.** *Fix an interior belief cut–off $u \in (\underline{\theta}_0, \overline{\theta}_0)$. Let*

$$m^{\text{req}}(u) := \mu^{-1}(u), \qquad \Lambda(\theta; u) := \alpha\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Assume that for this u the map $\theta \mapsto \Lambda(\theta; u)$ is strictly single–crossing, so the equation $\Lambda(\theta; u) = 0$ has a unique solution $\theta^*(u) \in (\underline{\theta}_0, u)$. Then $u \mapsto \theta^*(u)$ is continuously differentiable and

$$\frac{d\theta^*}{du} = -\frac{\Lambda_u\big(\theta^*(u); u\big)}{\Lambda_\theta\big(\theta^*(u); u\big)} \;=\; \frac{\big(\eta + \kappa\Delta(u)\big)\,\mu^{-1\prime}(u)}{\alpha + B'(\theta^*(u)) + \eta + \kappa\Delta(u)} \;>\; 0\,,$$

where $\Delta(u) := m^{\text{req}}(u) - \theta^*(u) > 0$. Hence the upper end of the pooling band θ^* moves strictly upward when the voting cut–off u is tightened.

Proof. By definition of $\theta^*(u)$, we have the identity

$$\Lambda\big(\theta^*(u); u\big) = 0 \qquad \text{for all interior } u.$$

[6] If $m^{\text{req}}(u) = \theta$, C may be kinked along the diagonal (allowed by (S1)); the one-sided derivatives obtained from the two wedges deliver the corresponding signs. Differentiability at the boundary is not needed for the comparative-statics results used later.

Because $u \in (\underline{\theta}_0, \overline{\theta}_0)$ and μ is strictly increasing and continuous, $m^{\text{req}}(u) = \mu^{-1}(u)$ is well defined and strictly increasing by (M1). Since $\theta^*(u) < u$, we have $\Delta(u) := m^{\text{req}}(u) - \theta^*(u) > 0$; i.e., the root lies on the *up–wedge*. By (S1)–(S4), C is C^1 off the diagonal $\{\Delta = 0\}$, so the partial derivatives below exist at $\big(\theta^*(u); u\big)$.

Differentiate the identity $\Lambda(\theta^*(u); u) = 0$ with respect to u. Using $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C\big(\mu^{-1}(u), \theta\big)$ gives

$$\Lambda_u(\theta; u) = -\, C_m\big(\mu^{-1}(u), \theta\big)\, \mu^{-1\prime}(u), \qquad \Lambda_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta\big(\mu^{-1}(u), \theta\big).$$

Evaluate at $\big(\theta^*(u); u\big)$. On the up–wedge ($\Delta > 0$), (S2)–(S4) imply

$$C_m\big(\mu^{-1}(u), \theta^*(u)\big) = \eta + \kappa\Delta(u) > 0, \qquad C_\theta\big(\mu^{-1}(u), \theta^*(u)\big) = -(\eta + \kappa\Delta(u)) < 0.$$

By (M1), $\mu^{-1\prime}(u) > 0$. Hence

$$\Lambda_u\big(\theta^*(u); u\big) = -\big(\eta + \kappa\Delta(u)\big)\, \mu^{-1\prime}(u) \;<\; 0,$$

and, using (P3) to get $B'(\theta) > 0$,

$$\Lambda_\theta\big(\theta^*(u); u\big) = \alpha + B'\big(\theta^*(u)\big) - C_\theta\big(\mu^{-1}(u), \theta^*(u)\big) = \alpha + B'\big(\theta^*(u)\big) + \eta + \kappa\Delta(u) \;>\; 0.$$

Since Λ is continuously differentiable in a neighborhood of the root and $\Lambda_\theta \neq 0$ there (strict single–crossing), the Implicit Function Theorem applies and yields

$$\frac{d\theta^*}{du} = -\frac{\Lambda_u\big(\theta^*(u); u\big)}{\Lambda_\theta\big(\theta^*(u); u\big)} = \frac{\big(\eta + \kappa\Delta(u)\big)\, \mu^{-1\prime}(u)}{\alpha + B'(\theta^*(u)) + \eta + \kappa\Delta(u)} \;>\; 0.$$

In particular, $0 < \frac{d\theta^*}{du} < \mu^{-1\prime}(u)$, with the bounds strict whenever $\Delta(u) > 0$ and $\eta > 0$. □

Lemma 4 (Pooling test)**.** *Let*

$$\Psi(u) \;:=\; \Lambda\big(\theta = \underline{\theta}_0\,;\, u\big),$$

i.e. the lying–profit function evaluated at the lowest quality in the support, $\theta = \underline{\theta}_0$. Then Ψ is continuous and weakly *decreasing in u; the inequality is strict whenever $m^{\text{req}}(u) > \underline{\theta}_0$. Therefore,*

$$\Pr(\textit{pooling}) > 0 \quad \Longleftrightarrow \quad \Psi(u) < 0.$$

Proof. Because $\Lambda(\theta; u)$ is continuous in both arguments, its restriction to $\theta = \underline{\theta}_0$ is also continuous, so Ψ is continuous. Lemma 1 shows $\partial\Lambda/\partial u \leq 0$ at every θ, with strict inequality whenever the effective lie is upward; in particular

$$\Psi'(u) \;=\; \frac{\partial\Lambda}{\partial u}(\underline{\theta}_0; u) \;=\; -\, C_m\big(m^{\text{req}}, \underline{\theta}_0\big)\, \mu^{-1\prime}(u).$$

Since $u = u > \underline{\theta}_0$, we have $m^{\text{req}} = \mu^{-1}(u) > \underline{\theta}_0$, hence $C_m > 0$ by 3.1.3; with $\mu^{-1\prime} > 0$ the derivative

is strictly negative.

Assume first that $\Psi(u) \geq 0$. Since $\Lambda(\theta; u)$ is strictly increasing in θ (single-crossing), the inequality $\Lambda(\underline{\theta}_0; u) \geq 0$ implies $\Lambda(\theta; u) \geq 0$ for every $\theta \geq \underline{\theta}_0$. The zero-crossing $\theta^*(u)$ therefore satisfies $\theta^* \leq \underline{\theta}_0$, so the interval $[\underline{\theta}_0, \theta^*)$ is empty and the probability of pooling, $F(\theta^*) - F(\underline{\theta}_0)$, equals 0.

Conversely, if $\Psi(u) < 0$ then $\Lambda(\underline{\theta}_0; u) < 0$. With $\Lambda(\cdot; u)$ strictly increasing, the function must cross zero at some $\theta^* > \underline{\theta}_0$, leaving a non-empty interval $[\underline{\theta}_0, \theta^*)$ on which $\Lambda < 0$; hence $F(\theta^*) - F(\underline{\theta}_0) > 0$.

Thus Pr(pooling) > 0 exactly when $\Psi(u) < 0$. □

4.3 General Existence and Uniqueness

As pointed out by Burkart et al. (2024), signaling problems often induce multiple equilibria (Shleifer and Vishny (1986)). To obtain a unique equilibrium outcome, We follow Burkart et al. (2024) in imposing the assumption that off-equilibrium beliefs satisfy the credible beliefs criterion of Grossman and Perry (1986), together with the log-concave density function of the CS's type as per Assumption 1. As per Bagnoli and Bergstrom (2005), this is a common assumption in signaling games with threshold strategies. In what follows, we define the existence of the unique equilibrum in these games— generalized to both voting rules.

Proposition 2 (Perfect–Sequential Equilibrium for a fixed belief cut-off)**.** *Fix any rule that induces the belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0]$. Define $m_{\text{req}}(u) := \mu^{-1}(u)$ and*

$$\Lambda(\theta; u) \;:=\; \alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big).$$

Let $\theta^*(u) := \sup\{\theta \in [\underline{\theta}_0, u) : \Lambda(\theta; u) < 0\}$. Minorities implement iff $\mu(m) \geq u$ (equivalently, iff $(1-\alpha)\mu(m) - \beta(m) \geq \underline{\kappa}^{(\lambda)}_{rule}$ by (V2)). Assume GP–credible beliefs off path. Throughout this proposition assume $0 < B'(\theta) < 1 - \alpha$ (P3).

(i) Existence. There exists a PSE with sender strategy

$$\sigma(\theta) = \begin{cases} \theta, & \theta < \theta^*(u) \quad \textit{(truth; rejected)}, \\ m_{\text{req}}(u), & \theta \in [\theta^*(u), u) \quad \textit{(cheapest lie; approved)}, \\ \theta, & \theta \geq u \quad \textit{(truth; approved)}. \end{cases}$$

(ii) Allocation uniqueness for a given u. For the same cut-off u, every PSE yields the same implement/reject allocation: the project is implemented if and only if $\theta \geq \theta^*(u)$.

(iii) Pooling vs. separation (endpoint test). Write

$$\Psi_\ell(u) \;:=\; \Lambda(\underline{\theta}_0; u), \qquad \Psi_h(u) \;:=\; \lim_{\theta \uparrow u} \Lambda(\theta; u) \;=\; \alpha u + B(u) - C\big(m^{\text{req}}(u), u\big).$$

Since $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing on $[\underline{\theta}_0, u]$, exactly one of the following obtains:

(A) Partial pooling: $\Psi_\ell(u) < 0 < \Psi_h(u) \implies \exists!\ \theta^*(u) \in (\underline{\theta}_0, u),\ F(u) - F(\theta^*(u)) > 0.$

(B) Full separation: $\Psi_h(u) \le 0 \implies \theta^*(u) = u$ *and the pooling set is empty.*

(C) Full pooling below u: $\Psi_\ell(u) \ge 0 \implies \theta^*(u) = \underline{\theta}_0$ *and all $\theta < u$ pool at $m_{\text{req}}(u)$.*

Proof. Preliminaries (voting and shape): By (M1), μ is continuous, strictly increasing, and invertible, so $m_{\text{req}}(u)$ exists and is unique and satisfies $\mu(m_{\text{req}}(u)) = u$. Define $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(m_{\text{req}}(u), \theta)$. Assumptions (S3)–(S4) and (P3) imply:

(a) $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing because

$$\Lambda_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta\big(m_{\text{req}}(u), \theta\big) \ \ge\ \alpha + B'(\theta) \ >\ 0;$$

(b) by Lemma 1, $\Lambda_u(\theta; u) = -C_m(m_{\text{req}}(u), \theta)\, \mu^{-1\prime}(u) \le 0$, with strict inequality whenever $m_{\text{req}}(u) > \theta$.

(i) Existence. We verify sender optimality type–by–type and then receivers' best responses.

Low types $\theta < \theta^*(u)$. By definition of $\theta^*(u)$, $\Lambda(\theta; u) < 0$. If a low type deviates to the least–cost approved message $m_{\text{req}}(u)$, her payoff is $\Lambda(\theta; u) < 0$, strictly below truthful rejection (0 by (S2)). Any other approved message $m > m_{\text{req}}(u)$ is strictly more costly because $C_m(m, \theta) > 0$ and is increasing on the up–wedge (S3), hence strictly worse. Any under–report $m < \theta$ is rejected since μ is increasing and $m < m_{\text{req}}(u)$ implies $\mu(m) < u$; moreover $C(m, \theta) > C(\theta, \theta) = 0$ by (S2)–(S3). Thus truth maximizes payoff for $\theta < \theta^*(u)$.

Persuaders $\theta \in [\theta^*(u), u)$. Truth would be rejected ($\mu(\theta) < u$), delivering 0. Sending $m_{\text{req}}(u)$ guarantees approval with payoff $\Lambda(\theta; u) \ge 0$. Among approved messages, $m_{\text{req}}(u)$ minimizes signaling cost by the same up–wedge argument as above, hence is strictly optimal.

High types $\theta \ge u$. Under the candidate strategy, high types separate truthfully on $[u, \overline{\theta}_0]$. By Bayes' rule along this separating segment, $\mu(m = \theta) = \theta \ge u$, so truth is approved at zero cost (S2). Any $m > \theta$ raises cost (no incremental benefit), and any $m < \theta$ either remains approved yet adds strictly positive cost on the down–wedge (S3), or falls below u and is rejected. Hence $m = \theta$ is optimal.

Receiver best responses and beliefs. Given σ above, the induced on–path messages are (i) $m = \theta$ for $\theta \ge u$ and (ii) $m = m_{\text{req}}(u)$ for $\theta \in [\theta^*(u), u)$. On path, Bayes' rule yields $\mu(\theta) = \theta \ge u$ for the separating region and $\mu(m_{\text{req}}(u)) = u$ for the pooling message, so minorities approve exactly when $\mu(m) \ge u$, i.e. when $m \ge m_{\text{req}}(u)$. Off path we adopt GP–credible beliefs; these do not profitably relax the cut–off for any deviator given the arguments above. Therefore the stated profile is a PSE.

(ii) Allocation uniqueness for fixed u. Let $(\hat{\sigma}, \widehat{\text{beliefs}})$ be any PSE with the same cut–off u. We

show the implementation set is $\{\theta \geq \theta^*(u)\}$.

Claim A. No $\theta < \theta^*(u)$ can be implemented. If $\hat{\sigma}(\theta)$ induces approval, then among approved messages the least costly is $m_{\text{req}}(u)$ (by S3), giving payoff $\Lambda(\theta; u) < 0$, strictly below truthful rejection 0.

Claim B. Every $\theta > \theta^*(u)$ must be implemented. If $\hat{\sigma}(\theta)$ is rejected, deviating to $m_{\text{req}}(u)$ yields $\Lambda(\theta; u) > 0$, a strict improvement. If $\hat{\sigma}(\theta)$ induces approval via some $m \neq \theta$, then for $\theta \geq u$ truth is approved at zero cost and strictly dominates; for $\theta \in (\theta^*(u), u)$ the least–cost approved message is $m_{\text{req}}(u)$, so any other approved m yields strictly lower payoff. Thus all $\theta > \theta^*(u)$ are implemented.

At $\theta = \theta^*(u)$, $\Lambda(\theta^*(u); u) = 0$; the type is indifferent between truthful rejection and sending $m_{\text{req}}(u)$, and either tie–break leaves all other allocations unchanged. Combining the claims gives the stated allocation uniqueness.

(iii) Pooling vs. separation. Set $\Psi_\ell(u) = \Lambda(\underline{\theta}_0; u)$ and $\Psi_h(u) = \lim_{\theta \uparrow u} \Lambda(\theta; u) = \alpha u + B(u) - C(m_{\text{req}}(u), u)$. Since $\Lambda(\cdot; u)$ is continuous and strictly increasing in θ on $[\underline{\theta}_0, u]$, the three cases follow by the intermediate value theorem: $\Psi_\ell(u) < 0 < \Psi_h(u)$ yields a unique $\theta^*(u) \in (\underline{\theta}_0, u)$ (partial pooling); $\Psi_h(u) \leq 0$ gives $\theta^*(u) = u$ (full separation); $\Psi_\ell(u) \geq 0$ gives $\theta^*(u) = \underline{\theta}_0$ (full pooling below u). □

In general, the economic intuition of Proposition 2 reflects the general problem of adverse selection where signaling is costly. In particular, because sending signals are costly, controllers with very low θ will often find it profitable to remain silent—paying the requisite cost to induce approval of the project yields strictly less than simply saying nothing and getting zero. On the other hand, the nature of the CS's utility function (C1) suggests that the net benefits from approval rise linearly with θ, while the convex costs from upward misreporting imply that θ does not rise as rapidly for controllers with very high types. Accordingly, these controllers will always refuse to stay in a rejected pool. As we will demonstrate later, this may generalize to the possibility that all controllers may refuse to stay in a rejected pool.

Proposition 3 (Comparative statics of pooling mass at a *fixed* belief cut–off (transferable B; talk–up))**.** *Maintain* (P1)–(P3), (S1)–(S4), *and* (M1) *in the payoff–dominant (talk–up) regime* $0 < B'(\theta) < 1 - \alpha$. *Fix a voting environment summarized by an interior belief cut–off* $u \leq 0$ *(treated as* exogenous and fixed *throughout this proposition)*[7] *and let* $\theta^*(u) < 0$ *be the unique root of* $\Lambda(\theta; u) =$

[7] If the belief cut–off u is *endogenous* to the parameter of interest via (V2)–(V3) (e.g., u depends on α or on the level of B through β), then the total derivative acquires an additional *bar–movement* term. Writing $P(u) = \int_{\theta^*(u)}^{u} f(\theta)\, d\theta$, the variable–limits Leibniz rule gives

$$P_u(u) = f(u) \;-\; f\big(\theta^*(u)\big)\, \theta^{*\prime}(u) \qquad \text{with} \qquad \theta^{*\prime}(u) = \frac{C_m\big(m_{\text{req}}(u), \theta^*(u)\big)\, (\mu^{-1})'(u)}{\alpha + B'\big(\theta^*(u)\big) \;-\; C_\theta\big(m_{\text{req}}(u), \theta^*(u)\big)} \;>\; 0,$$

so for any parameter p one has

$$\frac{dP}{dp} = P_u(u)\, \frac{du}{dp} \;-\; f\big(\theta^*(u)\big)\, \frac{\partial \theta^*}{\partial p}\Big|_u,$$

0 *on* $(\underline{\theta}_0, u)$*, where*

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big), \qquad m_{\text{req}}(u) = \mu^{-1}(u).$$

Define the pooling mass *(probability) as the mass of types who send* $m_{\text{req}}(u)$*:*

$$P(u) \;:=\; F(u) \;-\; F\big(\theta^*(u)\big).$$

Consider the following payoff–side *parameter changes (holding the voting environment, and hence* u *and* $m_{\text{req}}(u)$*, fixed):*

(a) Upward quality shift: an additive shift $\Delta > 0$ *in fundamentals so that the lying–profit becomes* $\Lambda_\Delta(\theta; u) := \alpha(\theta + \Delta) + B(\theta + \Delta) - C\big(m_{\text{req}}(u), \theta + \Delta\big)$*;*
(b) Higher ownership: $\alpha \mapsto \alpha + \Delta\alpha$*, with* $\Delta\alpha > 0$*;*
(c) Upward level *shift in private benefits:* $B(\theta) \mapsto B(\theta) + \Delta B$*, with* $\Delta B > 0$*.*

Then, at the fixed cut–off u*,*

$$\begin{aligned} \textit{(a) Upward quality } (\Delta > 0) :&\quad P \textit{ strictly} \text{ increases,} \\ \textit{(b) Higher ownership } (\Delta\alpha > 0) :&\quad P \textit{ strictly} \text{ decreases,} \\ \textit{(c) Higher private benefits } (\Delta B > 0) :&\quad P \textit{ strictly} \text{ increases.} \end{aligned}$$

Proof. All derivatives are computed at $\theta = \theta^*(u)$ with u fixed. By strict single–crossing on the up–wedge (see Lemma 2), $\Lambda_\theta(\theta; u) > 0$ on $(\underline{\theta}_0, u]$; hence, by the Implicit Function Theorem,

$$\frac{d\theta^*}{d\,\text{param}} \;=\; -\,\frac{\Lambda_{\text{param}}}{\Lambda_\theta} \qquad \text{evaluated at } \theta = \theta^*(u).$$

Because $P(u) = F(u) - F\big(\theta^*(u)\big)$ with u fixed,

$$\frac{dP}{d\,\text{param}} \;=\; -\,f\big(\theta^*(u)\big)\,\frac{d\theta^*}{d\,\text{param}},$$

and by (P2) the density satisfies $f > 0$ on the support. We compute Λ_{param} case by case.

(a) Upward quality $\Delta > 0$. Here Δ enters *only* as a shift of the type argument in each payoff term, so by the chain rule

$$\Lambda_\Delta(\theta; u) = \alpha + B'(\theta + \Delta) - C_\theta\big(m_{\text{req}}(u), \theta + \Delta\big).$$

where the second term is the *payoff–side* effect computed in the proposition (the "fixed–u" piece) and the first term is the *rule–side* effect through bar movement. Under additional structure (e.g., uniform prior with $\mu(m) = m$), $P_u(u) > 0$ along any pooling corridor, so the sign of dP/dp combines a positive bar–movement contribution with the payoff–side term; in general, the overall sign is parameter– and environment–dependent.

At $\Delta = 0$, $\Lambda_\Delta(\theta; u) = \alpha + B'(\theta) - C_\theta(m_{\text{req}}(u), \theta) = \Lambda_\theta(\theta; u)$. Thus, at the root,

$$\frac{d\theta^*}{d\Delta} = -\frac{\Lambda_\Delta}{\Lambda_\theta} = -1 \quad \Longrightarrow \quad \frac{dP}{d\Delta} = -f(\theta^*)\,\frac{d\theta^*}{d\Delta} = f(\theta^*) \;>\; 0.$$

(b) Ownership α. Differentiating $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(\cdot)$ with respect to α (holding u fixed) gives $\Lambda_\alpha(\theta; u) = \theta$. Evaluated at $\theta = \theta^*(u) < 0$ (since $u \leq 0$ and the root lies below u in the talk–up regime), we have $\Lambda_\alpha < 0$, hence

$$\frac{d\theta^*}{d\alpha} = -\frac{\Lambda_\alpha}{\Lambda_\theta} > 0 \quad \Longrightarrow \quad \frac{dP}{d\alpha} = -f(\theta^*)\,\frac{d\theta^*}{d\alpha} \;<\; 0.$$

(c) Level shift in B. A constant increase $B(\theta) \mapsto B(\theta) + \Delta B$ adds one–for–one to Λ, i.e. $\Lambda_{\Delta B} = +1$. Therefore

$$\frac{d\theta^*}{d(\Delta B)} = -\frac{1}{\Lambda_\theta} < 0 \quad \Longrightarrow \quad \frac{dP}{d(\Delta B)} = -f(\theta^*)\,\frac{d\theta^*}{d(\Delta B)} \;>\; 0.$$

All inequalities are strict because $f(\theta^*) > 0$ by (P2) and $\Lambda_\theta > 0$ by strict single–crossing (Lemma 2). □

Hold the belief cut–off u fixed (so the "bar" minorities apply does not move). When projects are uniformly stronger ($\Delta > 0$), the approval prize $\alpha(\theta + \Delta) + B(\theta + \Delta)$ increases one-for-one at every type while the cost of the *same* minimal lie to $m_{\text{req}}(u)$ is unchanged in u; marginal types just below the bar now find it worthwhile to talk up. The indifferent type $\theta^*(u)$ therefore shifts *down*, widening $[\theta^*(u), u]$ and raising the pooling mass $P(u) = F(u) - F(\theta^*(u))$.

Raising ownership ($\alpha \uparrow$) makes the controller internalize more of the project payoff. Along the relevant (weakly negative) part of the support, this *reduces* the net prize from approval (because $\alpha\theta$ is more negative for $\theta < 0$), so fewer borderline types are willing to pay the signaling cost to clear the same bar; the indifferent type moves *up* and the pooling band shrinks, lowering $P(u)$.

A uniform upward shift in private benefits ($B \uparrow$) adds a constant to the prize conditional on approval without making the message any more persuasive under the fixed threshold. With a larger prize, more near-bar types choose the minimal persuasive message; $\theta^*(u)$ moves *down*, the pooling band widens, and $P(u)$ rises.

5 Pooling Regions under Different Voting Regimes: Baseline

5.1 Majority-of-Minority versus a Lenient Super-Majority

The following propositions demonstrate a comparison between MoM and Lenient SM, where the requisite quota, π , is less than 0.5. We first show that both rules induce partial pooling under a very general assumption (Assumption (3)), before showing pooling mass comparisons between MoM and Lenient SM.

5.1.1 Existence of Partial Pooling

Assumption 3 (Local strengthened CPC at the bar)**.** *Fix an interior belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0)$ induced by the rule. Let $m^{\text{req}}(u) := \mu^{-1}(u)$. Assume*

$$\alpha\, u + B(u) \;>\; C\big(m^{\text{req}}(u), u\big). \tag{CPC$^+$}$$

Proposition 4 (Partial pooling under local CPC$^+$ for MoM and lenient SM)**.** *Maintain* P1–P3, S1–3.1.3, *and* M1 *in the payoff–dominant regime $0 < B'(\theta) < 1 - \alpha$, with $B(\underline{\theta}_0) = 0$. For a rule–specific interior belief cut–off $u \in (\underline{\theta}_0, \overline{\theta}_0)$ (either $u = u_{\text{MoM}}$ or $u = u_{\text{SM}}(\pi)$ with $\pi < \frac{1}{2}$), let $m^{\text{req}}(u) := \mu^{-1}(u)$ and define*

$$\Lambda(\theta; u) \;=\; \alpha\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Assume the local strengthened participation condition at the bar in Assumption CPC$^+$. Then for each rule (MoM or any lenient SM) there exists a unique $\theta^*(u) \in (\underline{\theta}_0, u)$ and the fixed-u PSE allocation *is partially pooling: truthful rejection on $[\underline{\theta}_0, \theta^*(u))$, pooling at $m^{\text{req}}(u)$ on $[\theta^*(u), u)$, and truthful approval on $[u, \overline{\theta}_0]$. Allocation uniqueness for a given u follows from Proposition 2*(ii).

Proof. We first prove that the lower endpoint of $\Lambda(\underline{\theta}_0; u)$ is negative. By (S2)–(S4) we have $C(\cdot, \theta) \geq 0$ with equality only at $m = \theta$, and by normalization $B(\underline{\theta}_0) = 0$. Hence

$$\Lambda(\underline{\theta}_0; u) = \alpha\, \underline{\theta}_0 + B(\underline{\theta}_0) - C\big(m^{\text{req}}(u), \underline{\theta}_0\big) \;\leq\; \alpha\, \underline{\theta}_0 \;<\; 0.$$

Next, we prove that the upper endpoint of $\Lambda(\underline{\theta}_0; u)$ is positive under local CPC$^+$. Define $\Psi_h(u) := \lim_{\theta \uparrow u} \Lambda(\theta; u) = \alpha u + B(u) - C\big(m^{\text{req}}(u), u\big)$. By Assumption CPC$^+$, $\Psi_h(u) > 0$.

By Lemma 2, the map $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing on $[\underline{\theta}_0, u]$.

Given the upper and lower endpoints of $\Lambda(\underline{\theta}_0; u)$ and Lemma 2, by the intermediate value theorem, we have a unique zero $\theta^*(u) \in (\underline{\theta}_0, u)$ with $\Lambda(\theta^*(u); u) = 0$. Because $f > 0$ on the support (Assumption 1), the pooling mass $F(u) - F(\theta^*(u))$ is strictly positive. Proposition 2(i)–(iii) then yields the fixed-u PSE allocation: truthful rejection on $[\underline{\theta}_0, \theta^*(u))$, pooling at $m^{\text{req}}(u)$ on $[\theta^*(u), u)$,

and truthful approval on $[u, \overline{\theta}_0]$. Allocation uniqueness for this u follows from Proposition 2(ii). □

Intuitively, fix any interior belief cut–off u. If the local strengthened CPC at the bar ($\alpha u + B(u) > C\big(\mu^{-1}(u), u\big)$) holds, then the fixed–$u$ equilibrium exhibits *partial pooling*: some types just below u shade up to the minimal persuasive message and pass, while higher types separate truthfully. If this inequality fails—e.g., under a lenient SM that drives u sufficiently left so the approval gain shrinks while $C\big(\mu^{-1}(u), u\big) > 0$ (since $\mu^{-1}(u) \neq u$ in general)—the entire pooling region collapses and the fixed–u equilibrium is *separating* at that bar.

5.1.2 Comparative Statics of Pooling Regions: Lenient SM vs MoM

Proposition 5 (Under uniform f and $\mu(m) = m$, MoM yields a larger pooling mass than any lenient SM)**.** *Maintain* P1–P3, S1–S4, *and* M1 *in the payoff–dominant (talk–up) regime. Assume the prior is uniform on $[a, b]$ with density $f(\theta) = \frac{1}{b-a}$ and $\mu(m) = m$, so $m^{\text{req}}(u) = u$. For each interior bar $u \in (a, b)$, define*

$$\Lambda(\theta; u) := \alpha\,\theta + B(\theta) - C(u, \theta), \qquad \theta^*(u) := \sup\{\theta < u \,:\, \Lambda(\theta; u) < 0\},$$

and pooling mass

$$P(u) \;:=\; F(u) - F\big(\theta^*(u)\big) \;=\; \int_{\theta^*(u)}^{u} f(\theta)\, d\theta \;=\; \frac{u - \theta^*(u)}{b - a}.$$

Given that a lenient SM induces a lower bar than MoM (i.e. $u_{\text{SM}}(\pi) < u_{\text{MoM}}$ for $\pi < \frac{1}{2}$), then

$$P\big(u_{\text{SM}}(\pi)\big) \;<\; P\big(u_{\text{MoM}}\big)$$

i.e., MoM produces a strictly larger pooling probability than any lenient SM.

Proof. By Lemma 2 there is a unique $\theta^*(u) \in (a, u)$ with $\Lambda(\theta^*(u); u) = 0$. Since $\mu(m) = m$, the minimal persuasive message at bar u is $m^{\text{req}}(u) = u$, so at the boundary the lie size is $\Delta(u) := u - \theta^*(u) > 0$ (up–wedge). Under gap costs (S4) we have $C_\theta(u, \theta) = -C_m(u, \theta)$ and, on the up–wedge, $C_m(u, \theta) = \eta + \kappa\Delta > 0$; in particular,

$$C_m\big(u, \theta^*(u)\big) > 0.$$

Moreover, by (P3) we have $0 < B'(\theta) < 1 - \alpha$, hence

$$\alpha + B'\big(\theta^*(u)\big) \;>\; 0.$$

Deriving the expression $\theta^{*\prime}(u)$ from Lemma 3 and using $\mu(m) = m$ with $C_\theta = -C_m$ gives:

$$\theta^{*\prime}(u) = -\frac{\Lambda_u}{\Lambda_\theta} = \frac{C_m\big(u, \theta^*(u)\big)}{\alpha + B'\big(\theta^*(u)\big) + C_m\big(u, \theta^*(u)\big)}.$$

Because both the numerator $C_m(u, \theta^*(u))$ and the added term $\alpha + B'(\theta^*(u))$ in the denominator are strictly positive, it follows that

$$0 \;<\; \theta^{*\prime}(u) \;<\; 1.$$

Write the pooling mass as

$$P(u) = \int_{\theta^*(u)}^{u} f(\theta)\, d\theta.$$

By the variable–limits Leibniz rule, the derivative is

$$P'(u) = f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u).$$

where we used the chain rule on the moving lower limit: $\frac{d}{du}\int_{\theta^*(u)}^{\cdot} f(\theta)\, d\theta = f(\theta^*(u))\, \theta^{*\prime}(u)$.

Equivalently, comparing to zero shows that the sign of $P'(u)$ is governed by a bar–boundary race:

$$\theta^{*\prime}(u) \lesseqgtr \frac{f(u)}{f(\theta^*(u))} \iff P'(u) \gtreqless 0$$

i.e., the pooling mass rises when the boundary moves more slowly than the density ratio, and falls when it moves faster.

Under the uniform prior, $f(u) = f(\theta^*(u)) = \frac{1}{b-a}$, so $\frac{f(u)}{f(\theta^*(u))} = 1$. Given that $0 < \theta^{*\prime}(u) < 1$, hence $\theta^{*\prime}(u) < \frac{f(u)}{f(\theta^*(u))}$ and, by evaluating the expression $\theta^{*\prime}(u) \lesseqgtr \frac{f(u)}{f(\theta^*(u))}$, $P'(u) > 0$ on any corridor with partial pooling.

In the talk–up regime, a lenient SM lowers the bar relative to MoM: $u_{\mathrm{SM}}(\pi) < u_{\mathrm{MoM}}$ for $\pi < \frac{1}{2}$. Integrating $P'(u) > 0$ over $[u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}]$ gives $P(u_{\mathrm{MoM}}) - P(u_{\mathrm{SM}}(\pi)) = \int_{u_{\mathrm{SM}}(\pi)}^{u_{\mathrm{MoM}}} P'(u)\, du > 0$, so $P(u_{\mathrm{SM}}(\pi)) < P(u_{\mathrm{MoM}})$. □

5.1.3 Sufficient Conditions for Pooling Region Comparative Statics (Up-Talk)

5.1.3.1 Bar-Boundary Comparisons

As the sign of the $P'(u)$ depends on the crucially on the "bar-boundary" comparison outlined in Proposition 5 (Milgrom and Shannon (1994)), it is instructive to consider a more general form of the inequality:

$$\theta^{*\prime}(u) \lesseqgtr \frac{f(u)}{f(\theta^*(u))}$$

where

$$\begin{aligned}
\theta^{*\prime}(u) &= -\frac{\Lambda_u(\theta^*(u); u)}{\Lambda_\theta(\theta^*(u); u)} \\
&= \frac{C_m\big(m^{\text{req}}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)}{\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(m^{\text{req}}(u), \theta^*(u)\big)} \\
&= \frac{C_m\big(\mu^{-1}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)}{\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(\mu^{-1}(u), \theta^*(u)\big)}
\end{aligned}$$

from Lemma 3 represents the behavioral responsiveness of the marginal type to a small change in the bar u, and:

$$r(u) := \frac{f(u)}{f(\theta^*(u))}$$

is the density ratio at the two endpoints of the pooling band, capturing how the raw probability mass at the bar compares to the mass "weight" at the moving boundary.

Both the terms $\frac{C_m\left(\mu^{-1}(u),\theta^*(u)\right)(\mu^{-1})'(u)}{\alpha+B'\left(\theta^*(u)\right)-C_\theta\left(\mu^{-1}(u),\theta^*(u)\right)}$ and $r(u)$ are economically meaningful. Insofar as the former term is concerned, consider the marginal type $\theta^*(u)$ who is just indifferent between telling the truth (rejected) and shading up to the bar u (approved). When the bar increases by $du > 0$, the required lie changes by $d\Delta = du - d\theta^*$. Holding the boundary fixed, this raises the marginal signaling burden by $C_m\big(\mu^{-1}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)\,du$ and therefore reduces the set of types willing to shade. The boundary then adjusts by $[\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(\mu^{-1}(u), \theta^*(u)\big)]\,d\theta^*$ to restore indifference. This adjustment provides two offsetting benefits for higher types: (i) the truthful payoff increases at slope $\alpha + B'(\theta^*(u))$, and (ii) because gap costs satisfy $C_\theta = -C_m$, raising θ^* shrinks the lie and lowers the signaling cost at rate $C_m(\mu^{-1}(u), \theta^*(u))$. Rewriting $\theta^{*\prime}(u)$:

$$\theta^{*\prime}(u) = -\frac{\Lambda_u}{\Lambda_\theta} = \frac{C_m\big(\mu^{-1}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)}{\alpha + B'\big(\theta^*(u)\big) + C_m\big(\mu^{-1}(u), \theta^*(u)\big)}$$

The *numerator*, $C_m\big(\mu^{-1}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)$, is the marginal cost pressure associated with the bar move du: it measures how much more expensive shading becomes per unit tightening. The *denominator*, $\alpha + B'\big(\theta^*(u)\big) + C_m\big(\mu^{-1}(u), \theta^*(u)\big)$, is the total marginal benefit associated with the boundary move $d\theta^*$: the first term $\alpha + B'$ is the truthful-payoff slope (higher types gain more from truth), and the $+\,C_m$ term captures the cost relief from reducing the lie for higher types (since $C_\theta = -C_m$ under gap costs).

The second term, $r(u) := f(u)/f(\theta^*(u))$, is a purely mechanical "thickness" comparison of the prior at the two endpoints of the pooling band $[\theta^*(u), u]$. A small increase in the bar by du adds raw probability mass at the bar at rate $f(u)\,du$, while a same-sized movement of the lower endpoint would release mass at rate $f(\theta^*(u))\,du$. Thus $r(u)$ benchmarks how much mass the bar picks up per unit shift relative to what the boundary would release per unit shift, *holding behavior fixed*. Under a log–concave (unimodal) density with monotone shoulders, the benchmark depends

on where the band sits: on the *right shoulder* (where $u > \theta^*(u)$ and f is weakly decreasing), we have $f(u) \leq f(\theta^*(u))$ so $r(u) \leq 1$, meaning the bar lies in a thinner tail and per–unit movements mechanically contribute less mass than at the boundary; on the *left shoulder* (where f is weakly increasing), $f(u) \geq f(\theta^*(u))$ so $r(u) \geq 1$, meaning the bar sits in a thicker region and per–unit movements mechanically contribute more mass than at the boundary. This density ratio is the benchmark in the bar–boundary comparison $P'(u) = f(u) - f(\theta^*(u))\, \theta^{*\prime}(u)$: it isolates the purely probabilistic (non-behavioral) side of how much mass a marginal shift of the bar can add relative to an equal shift of the band's lower endpoint.

5.1.3.2 Intuition behind Proposition 5

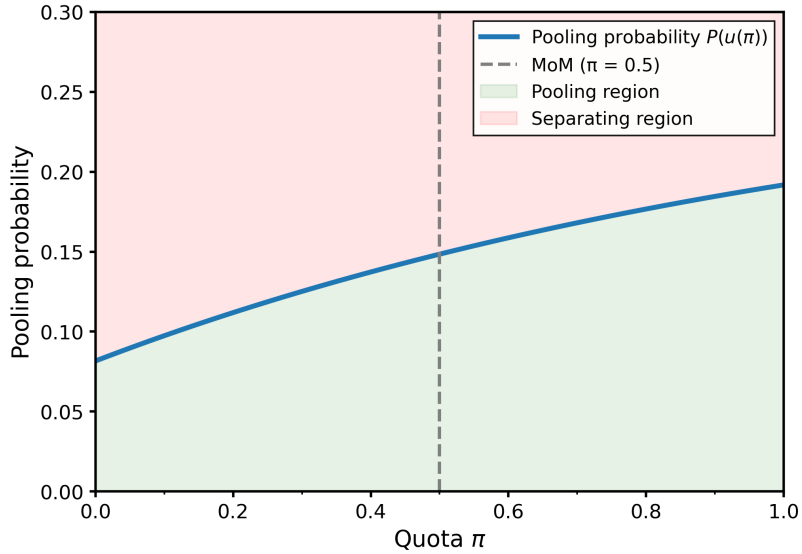
In the special case of Proposition 5 where $\mu(m) = m$, $\mu^{-1}(u) = u$, and $(\mu^{-1})'(u) = 1$. Thus, $\theta^{*\prime}(u)$ may be rewritten as $\theta^{*\prime}(u) = -\frac{\Lambda_u}{\Lambda_\theta} = \frac{C_m(u,\theta^*)}{\alpha + B'(\theta^*) + C_m(u,\theta^*)}$. Since $C_m(u, \theta^*) > 0$ and $\alpha + B'(\theta^*) > 0$ by derivation (Proposition 5), the denominator strictly exceeds the numerator: $0 < \theta^{*\prime}(u) < 1$. In words, the behavioral response moves the boundary by less than one-for-one when the bar shifts.

Turning to the mechanical side, under a uniform prior the density is flat, so raising the bar by du adds exactly one unit of mass per unit distance at the bar while an equal movement at the boundary releases mass at the same unit rate; equivalently, $r(u) = f(u)/f(\theta^*(u)) = 1$. Combining the two observations, $\theta^{*\prime}(u) < r(u)$: the boundary lags a one-for-one mass shift. Consequently, the pooling interval $[\theta^*(u), u]$ widens and the pooling mass $P(u)$ increases as we move from a lenient SM (lower u) to MoM (higher u), consistent with the bar–boundary comparison $P'(u) = f(u) - f(\theta^*(u))\, \theta^{*\prime}(u)$.

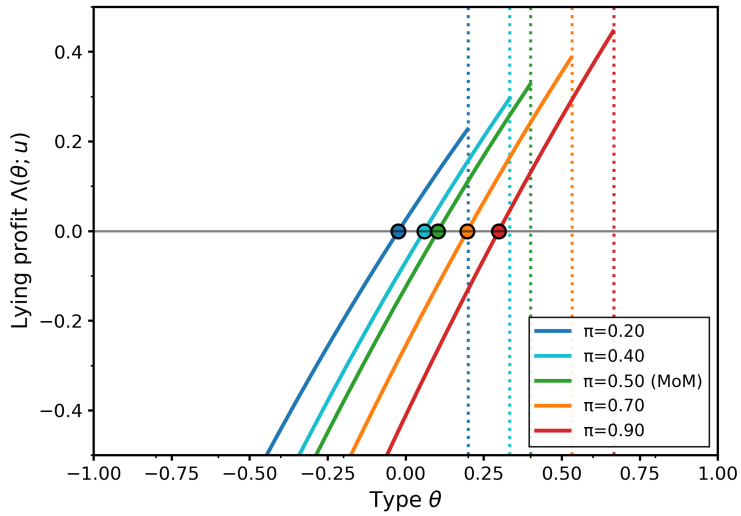
As shown in Figure 1, the bar advances faster than the boundary, so the gap $u - \theta^*(u)$ increases and the pooling mass $P(u)$ rises when moving from a lenient SM to MoM.

Figure 1: Pooling Probabilities and Lying-Profit Curves (Uniform f, Identity Beliefs)

(a) Panel A: Pooling Probabilities



(b) Panel B: Lying-Profit Curves



Panel A plots the simulated pooling probability $P(u(\pi))$ (solid blue) where the controller's cost is $C(m, \theta) = \eta|m - \theta| + \frac{\kappa}{2}(m - \theta)^2$; parameter values are $(\alpha, b_0, b_1, \lambda, K, \eta, \kappa) = (0.50, 0.12, 0.20, 0.10, 0.60, 0.20, 0.50)$. The grey dashed line marks the majority-of-the-minority threshold $\pi = 0.5$ and the green (pink) shading indicates the pooling (separating) region. Panel B shows the lying-profit function on the up-wedge, $\Lambda(\theta, u) = (\alpha + \beta)\theta - C(u)$, for $\pi \in \{0.20, 0.40, 0.50, 0.59, 0.61\}$; solid curves correspond to quotas that still permit pooling, dashed/dotted curves to quotas with complete separation, vertical dotted lines mark the induced belief cut-off $u = u_{\text{rule}}(\pi)$, and open markers indicate the type cutoff θ^* where $\Lambda(\theta, u) = 0$.

5.1.3.3 Density–Weighted Boundary Responsiveness: Microfoundations

The relative sizes of $r(u)$ and $\theta^{*\prime}(u)$ depend on the primitives—namely the shape of the prior f, the belief map μ, and the marginal costs and benefits of shading to the bar u (see Section 5.1.3.1). To obtain clean, tractable comparative statics across the pooling regions induced by different voting rules, we provide microfounded sufficient conditions under which the bar–boundary comparison is signable. The key additional feature we assume here is that of *sluggish beliefs*, captured by a local responsiveness parameter $\mu'(m_{\text{req}}(u)) = K \in (0, 1]$, so that minorities do not update one–for–one with shifts in the belief cut–off. This behavioral feature can be motivated by standard theories of noisy signaling, rational inattention, or endogenous non–participation, and it enters the boundary–slope expression multiplicatively through $(\mu^{-1})'(u) = 1/K$, delivering a transparent dial that links primitives to the density–weighted responsiveness test (Blume et al., 2007; Sims, 2003; Matějka and Tabellini, 2021; Palfrey and Rosenthal, 1983; Osborne et al., 2000; Levine and Palfrey, 2007).

We assume $\mu(m) = K\,m$ with $K \in (0, 1]$, the gap–cost specification from (S4), and affine private benefits $B(\theta) = b_0 + b_1\theta$. Under these primitives, $(\mu^{-1})'(u) = 1/K$. Substituting into the general boundary–slope formula from Lemma 3 yields the explicit expression

$$\theta^{*\prime}(u) = \frac{\eta + \kappa\,\Delta(u)}{\alpha + b_1 + \eta + \kappa\,\Delta(u)} \cdot \frac{1}{K}, \qquad K \in (0, 1], \;\; \Delta(u) > 0.$$

Hence $\theta^{*\prime}(u) \in \big(0,\, 1/K\big)$, it is strictly increasing in the required lie $\Delta(u)$, and

$$\lim_{\Delta \downarrow 0} \theta^{*\prime}(u) = \frac{\eta}{\alpha + b_1 + \eta} \cdot \frac{1}{K}.$$

Under log–concavity with a unique mode at u_{MoM}, the shoulder ratio $r(u) = f(u)/f(\theta^*(u))$ satisfies $r(u) \geq 1$ on the left corridor $u \in [u_{\text{SM}}(\pi), u_{\text{MoM}}]$ and $r(u) \leq 1$ on the right corridor $u \in [u_{\text{MoM}}, \bar{u}]$. Because $\theta^{*\prime}(u)$ is increasing in the required lie $\Delta(u)$, its *minimum* on any corridor occurs as $\Delta(u) \downarrow 0$. Hence a simple global bound at the corridor's thinnest lie delivers a *sufficient condition* for the density–weighted comparison:

$$\frac{\eta}{K\big(\alpha + b_1 + \eta\big)} \;\geq\; \sup_{u \text{ on the comparison path}} r(u)$$

Under this bound we have $\theta^{*\prime}(u) \geq r(u)$ for all interior u on the path, i.e. the generalized density–weighted boundary responsiveness Assumption 4 holds.[8]

Assumption 4 (Global density–weighted boundary responsiveness (GDWBR))**.** *In the payoff–dominant (talk–up) regime, for every interior bar u at which partial pooling holds and $\theta^*(u) \in$*

[8]For a uniform prior, $r(u) = 1$, so the sufficient condition reduces to $\theta^{*\prime}(u) \geq 1$ along the path.

$(\underline{\theta}_0, u)$, *impose the global inequality*

$$\theta^{*\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))}$$

for all u along the lenient corridor $u \in [u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}]$ and the strict corridor $u \in [u_{\mathrm{MoM}}, \bar{u}]$, while pooling persists.

Intuitively, when beliefs are sluggish ($\mu'(m_{\mathrm{req}}(u)) = K \in (0, 1]$) the boundary slope picks up the multiplier $(\mu^{-1})'(u) = 1/K$, so a given change in the bar requires a larger signal adjustment to restore indifference. Economically, slower updating by minorities makes near–threshold senders shade more aggressively, which pushes the marginal type (the boundary) to move faster. As K falls, the left–hand side $\eta/[K(\alpha + b_1 + \eta)]$ rises, making it easier to dominate the density ratio $\sup_{u \text{ on path}} r(u)$ and thus satisfy the global condition.

Proposition 6 (Lenient SM yields larger pooling mass than MoM under GDWBR)**.** *Maintain* P1–P3, S1–S4, *and* M1 *in the payoff–dominant (talk–up) regime, and let $u_{\mathrm{SM}}(\pi)$ and u_{MoM} be interior belief cut–offs with $u_{\mathrm{SM}}(\pi) < u_{\mathrm{MoM}}$ (lenient SM vs. MoM). Assume partial pooling survives for all u in the corridor $[u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}]$ and that the* Global Density–Weighted Boundary Responsiveness *(GDWBR) condition holds there:*

$$\theta^{*\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))} \qquad \textit{for all interior } u \in [u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}].$$

Then

$$P\big(u_{\mathrm{SM}}(\pi)\big) \;>\; P\big(u_{\mathrm{MoM}}\big)$$

i.e., any lenient SM produces a strictly larger pooling probability than MoM.

Proof. Fix $u \in [u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}]$. By Lemma 2 (single crossing on the up–wedge), there exists a unique boundary $\theta^*(u) \in (\underline{\theta}_0, u)$ such that $\Lambda(\theta^*(u); u) = 0$. Define the pooling mass below the bar by

$$P(u) \;:=\; \int_{\theta^*(u)}^{u} f(\theta)\, d\theta,$$

which is strictly positive because $f > 0$ on the support (Assumption 1) and $\theta^*(u) < u$. The boundary is differentiable in u on the corridor by the implicit–function theorem applied to $\Lambda(\theta; u) = 0$ with $\Lambda_\theta(\theta^*(u); u) > 0$ (again, invoking Lemma 2).

Differentiate $P(u) = \int_{\theta^*(u)}^{u} f(\theta)\, d\theta$ using the variable–limits Leibniz rule:

$$P'(u) = f(u) \cdot 1 \;-\; f\big(\theta^*(u)\big) \cdot \theta^{*\prime}(u) = f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u).$$

By GDWBR on the corridor,

$$\theta^{*\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))}.$$

Substitute this into $P'(u)$ to obtain

$$P'(u) \;\le\; f(u) - f(\theta^*(u)) \cdot \frac{f(u)}{f(\theta^*(u))}$$

with strict inequality on any subinterval where GDWBR is strict or f is not everywhere flat.

Integrate $P'(u) \le 0$ over $u \in [u_{\mathrm{SM}}(\pi), u_{\mathrm{MoM}}]$:

$$P\big(u_{\mathrm{MoM}}\big) - P\big(u_{\mathrm{SM}}(\pi)\big) = \int_{u_{\mathrm{SM}}(\pi)}^{u_{\mathrm{MoM}}} P'(u)\, du \;\le\; 0,$$

with strict inequality under the nondegeneracy noted in $P'(u) \;\le\; f(u) - f(\theta^*(u)) \cdot \frac{f(u)}{f(\theta^*(u))}$. Rearranging yields $P(u_{\mathrm{SM}}(\pi)) > P(u_{\mathrm{MoM}})$, as claimed. □

The intuition behind Proposition 6 mirrors that of Proposition 5. Tightening the belief cut–off u (monotone in the rule) shifts the marginal type $\theta^*(u)$ who is just indifferent between truth (rejected) and shading up to u (approved). With sufficiently sluggish beliefs, minorities' updates are attenuated, so a higher bar requires a disproportionately larger signal. The marginal benefit to slightly higher types from shading then outweighs the marginal cost, making the boundary move quickly—indeed fast enough that the behavioral response (captured by $\theta^{*\prime}(u)$) exceeds the purely mechanical mass effect at the bar (the density ratio $f(u)/f(\theta^*(u))$). Under this global condition, a tighter u reduces pooling mass ($P'(u) \le 0$), so a lenient SM (lower u) necessarily yields more pooling than MoM.

5.2 Majority-of-Minority versus a Stringent Super-Majority

The following propositions demonstrate a comparison between MoM and Stringent SM, where the requisite quota, π , is more than 0.5. We first show that the stringent SM may induce complete separation for a sufficiently high quota, $\bar{\pi}$, before showing that the MoM induces relatively less pooling than the stringent SM rule under standard priors.

Proposition 7 (Stringent SM: partial pooling below a critical quota, full separation above)**.** *Maintain* P1–P3*,* S1–S4*,* M1*, and the tail–growth assumption on talk-up costs. Fix a* stringent *quota* $\pi \in (\frac{1}{2}, 1]$ *and let* $u_{\mathrm{SM}}(\pi)$ *denote the induced (rule–specific) belief cut–off defined by* (V2)*; assume it is interior whenever invoked. Write* $m^{\mathrm{req}}(u) := \mu^{-1}(u)$, $\Lambda(\theta; u) := \alpha\theta + B(\theta) - C\big(m^{\mathrm{req}}(u), \theta\big)$*, and* $\theta^*(u) := \sup\{\theta < u : \Lambda(\theta; u) < 0\}$.

(i) Interior mapping. For any stringent π *with interior* $u_{\mathrm{SM}}(\pi)$*, the interior cut–off is unique. Moreover, on its domain, the map* $\pi \mapsto u_{\mathrm{SM}}(\pi)$ *is strictly increasing; if* $\beta \circ \mu^{-1}$ *is* C^1*, then* u_{SM} *is* C^1 *with*

$$\frac{du_{\mathrm{SM}}}{d\pi} \;=\; \frac{2/\lambda}{(1-\alpha) - \beta_u\big(u_{\mathrm{SM}}(\pi)\big)} \;>\; 0.$$

(ii) Dichotomy for stringent SM under CPC$^+$. *There exists* $\bar{\pi} \in (\frac{1}{2}, 1]$ *such that:*

$$\begin{cases} \text{If } \frac{1}{2} < \pi < \bar{\pi}, \text{ CPC}^+ \text{ holds at } u_{\mathrm{SM}}(\pi) \implies \underline{\theta}_0 < \theta^*\big(u_{\mathrm{SM}}(\pi)\big) < u_{\mathrm{SM}}(\pi) \text{ (partial pooling);} \\ \text{If } \pi \geq \bar{\pi}, \implies \theta^*\big(u_{\mathrm{SM}}(\pi)\big) = u_{\mathrm{SM}}(\pi) \text{ (full separation).} \end{cases}$$

If, in addition, $L(u) := u - \theta^*(u)$ *is strictly decreasing on the stringent corridor, then* $\bar{\pi}$ *is unique.*

Proof. (i) Interior mapping. By (V2), an interior cut–off u for a given π solves a strictly increasing equation in u (since $u \mapsto \beta(\mu^{-1}(u))$ is strictly increasing), which pins down uniqueness of $u_{\mathrm{SM}}(\pi)$. Differentiating the (V2) identity yields the displayed derivative when $\beta \circ \mu^{-1}$ is C^1, and the denominator is positive by (P3); hence $\pi \mapsto u_{\mathrm{SM}}(\pi)$ is strictly increasing.

(ii) Dichotomy. Fix a stringent π with interior $u := u_{\mathrm{SM}}(\pi)$ and define

$$\Psi_\ell(u) := \Lambda(\underline{\theta}_0; u), \qquad \Psi_h(u) := \lim_{\theta \uparrow u} \Lambda(\theta; u) = \alpha u + B(u) - C\big(m^{\mathrm{req}}(u), u\big).$$

Since $u > \underline{\theta}_0$ we have $m^{\mathrm{req}}(u) = \mu^{-1}(u) > \underline{\theta}_0$ and thus $C(m^{\mathrm{req}}(u), \underline{\theta}_0) > 0$ by (S2)–(S4); with $B(\underline{\theta}_0) = 0$ and $\alpha\underline{\theta}_0 < 0$, we get $\Psi_\ell(u) < 0$. Under CPC$^+$ at the induced bar, $\Psi_h(u) > 0$. By Lemma 2 (single crossing on $[\underline{\theta}_0, u]$), there is a unique $\theta^*(u) \in (\underline{\theta}_0, u)$, and Proposition 2 yields partial pooling for this u.

Next, consider tightening the quota. By (M1′) (full range, defined elsewhere) and tail growth of the gap costs, as $u \uparrow \overline{\theta}_0$ we have $m^{\mathrm{req}}(u) = \mu^{-1}(u) \uparrow +\infty$, so for $\theta \uparrow u$ the gap $\Delta(u, \theta) = m^{\mathrm{req}}(u) - \theta \to \infty$ and $C\big(m^{\mathrm{req}}(u), \theta\big) \to \infty$; hence $\Psi_h(u) \to -\infty$. By continuity of Ψ_h in u, there exists at least one interior $\bar{u}$ with $\Psi_h(\bar{u}) = 0$ (and $\Psi_h(u) > 0$ for $u < \bar{u}$ in a neighborhood, by the preceding argument).

Define $\bar{\pi}$ as the quota whose induced cut–off equals $\bar{u}$, i.e. $\bar{\pi}$ solves (V2) at $u = \bar{u}$. Part (i) implies $\pi \mapsto u_{\mathrm{SM}}(\pi)$ is strictly increasing; therefore: - If $\frac{1}{2} < \pi < \bar{\pi}$, then $u_{\mathrm{SM}}(\pi) < \bar{u}$ and $\Psi_h\big(u_{\mathrm{SM}}(\pi)\big) > 0$, yielding partial pooling. - If $\pi \geq \bar{\pi}$, then $u_{\mathrm{SM}}(\pi) \geq \bar{u}$ and $\Psi_h\big(u_{\mathrm{SM}}(\pi)\big) \leq 0$, so by Proposition 2(iii) the zero satisfies $\theta^*\big(u_{\mathrm{SM}}(\pi)\big) = u_{\mathrm{SM}}(\pi)$ (full separation).

If, moreover, $L(u) := u - \theta^*(u)$ is strictly decreasing along the stringent corridor, then $L(u_{\mathrm{SM}}(\pi))$ is strictly decreasing in π and the threshold is unique. □

The intuition behind Proposition 7 is as follows: Tightening the super-majority quota raises the belief cut-off u required for approval. To clear a higher bar, the controller must send a larger persuasive message $m_{\mathrm{req}}(u) = \mu^{-1}(u)$, which increases the lie $m_{\mathrm{req}}(u) - \theta$ for types just below u. With tail-growth in signaling costs, the expense of sufficiently large lies eventually overwhelms the bounded approval prize $\alpha u + B(u)$ (types reside on a compact support). At that point—corresponding to a critical quota $\bar{\pi}$—even the marginal type will not misreport, so the equilibrium becomes fully separating. For less strict quotas, CPC$^+$ ensures that the prize at the bar still exceeds the minimal signaling cost, so a nonempty set of near-threshold types shade up and partial pooling obtains.

Proposition 8 (Under uniform f and $\mu(m) = m$, any *strict* SM yields a larger pooling mass than MoM (on the pooling corridor)). *Maintain* P1–P3, S1–S4, *and* M1 *in the payoff–dominant (talk–up) regime. Assume the prior is uniform on* $[a, b]$ *with density* $f(\theta) = \frac{1}{b-a}$ *and* $\mu(m) = m$, *so* $m^{\text{req}}(u) = u$. *For each interior bar* $u \in (a, b)$, *define*

$$\Lambda(\theta; u) := \alpha\,\theta + B(\theta) - C(u, \theta), \qquad \theta^*(u) := \sup\{\theta < u : \Lambda(\theta; u) < 0\},$$

and the pooling mass

$$P(u) := \int_{\theta^*(u)}^{u} f(\theta)\,d\theta = \frac{u - \theta^*(u)}{b - a}.$$

Let u_{MoM} *denote the MoM bar and let* $u_{\text{SM}}(\pi)$ *be the bar induced by a* strict *super-majority quota* $\pi > \frac{1}{2}$ *and* $u_{\text{SM}}(\pi) > u_{\text{MoM}}$. *Assume both bars are interior and that partial pooling holds for all* u *along the corridor* $[u_{\text{MoM}}, u_{\text{SM}}(\pi)]$. *Then*

$$P\big(u_{\text{SM}}(\pi)\big) > P\big(u_{\text{MoM}}\big)$$

i.e., a strict SM produces a strictly larger pooling probability than MoM on the pooling corridor.

Proof. By Lemma 2 there is a unique $\theta^*(u) \in (a, u)$ with $\Lambda(\theta^*(u); u) = 0$ when pooling holds. Since $\mu(m) = m$, the least–cost approved message is $m^{\text{req}}(u) = u$, so the lie at the boundary is $\Delta(u) := u - \theta^*(u) > 0$ (up–wedge). Gap costs (S4) give $C_\theta(u, \theta) = -C_m(u, \theta)$ and, on the up–wedge, $C_m(u, \theta) = \eta + \kappa\Delta > 0$; hence

$$C_m\big(u, \theta^*(u)\big) > 0.$$

With $0 < B'(\theta) < 1 - \alpha$ (P3), we have $\alpha + B'(\theta^*(u)) > 0$. Deriving the expression $\theta^{*\prime}(u)$ from Lemma 3 and using $\mu(m) = m$ with $C_\theta = -C_m$ gives:

$$\theta^{*\prime}(u) = -\frac{\Lambda_u}{\Lambda_\theta} = \frac{C_m\big(u, \theta^*(u)\big)}{\alpha + B'\big(\theta^*(u)\big) + C_m\big(u, \theta^*(u)\big)} \in (0, 1).$$

With $f \equiv \frac{1}{b-a}$ and $P(u) = \frac{u-\theta^*(u)}{b-a}$,

$$P'(u) = \frac{1 - \theta^{*\prime}(u)}{b - a} > 0,$$

since $0 < \theta^{*\prime}(u) < 1$ as established.

For a strict SM, the induced bar satisfies $u_{\text{SM}}(\pi) > u_{\text{MoM}}$. By the definition of $P'(u)$ and the corridor pooling assumption,

$$P\big(u_{\text{SM}}(\pi)\big) - P\big(u_{\text{MoM}}\big) = \int_{u_{\text{MoM}}}^{u_{\text{SM}}(\pi)} P'(u)\,du > 0.$$

This proves the claim. □

The intuition of Proposition 8 is similar to Proposition 5. A stricter super-majority raises the belief cut-off u, so to pass the vote the controller must send a higher persuasive message $m_{\text{req}}(u) = u$, which increases the required lie for types just below the bar. With gap costs, the marginal type's best-response boundary $\theta^*(u)$ moves *less than one-for-one* when the bar moves (formally $0 < \theta^{*\prime}(u) < 1$), so the lower endpoint of the pooling band lags behind the rising bar. Under a uniform prior this mechanical "mass at the bar" equals the "mass at the boundary," so the bar's faster movement relative to the boundary strictly widens the interval $[\theta^*(u), u]$. Hence, along any corridor where partial pooling persists, a strict SM (higher u) produces a larger pooling mass than MoM.

Proposition 9 (Strict SM yields smaller pooling mass than MoM under GDWBR)**.** *Maintain* P1–P3*,* S1–S4*, and* M1 *in the payoff–dominant (talk–up) regime, and let $u_{\text{SM}}(\pi)$ and u_{MoM} be interior belief cut–offs with $u_{\text{SM}}(\pi) > u_{\text{MoM}}$ (strict SM vs. MoM). Assume partial pooling survives for all u in the corridor $[u_{\text{MoM}},\ u_{\text{SM}}(\pi)]$ and that the* Global Density–Weighted Boundary Responsiveness *(GDWBR) condition holds there:*

$$\theta^{*\prime}(u) \ \geq\ \frac{f(u)}{f(\theta^*(u))} \qquad \textit{for all interior } u \in [u_{\text{MoM}},\ u_{\text{SM}}(\pi)].$$

Then

$$P\big(u_{\text{SM}}(\pi)\big) \ \leq\ P\big(u_{\text{MoM}}\big)$$

with strict inequality whenever GDWBR is strict on a subset of positive measure or f is not everywhere flat on the corridor. Hence, on the pooling corridor, any strict SM produces a (weakly) smaller pooling probability than MoM.

Proof. Fix $u \in [u_{\text{MoM}},\ u_{\text{SM}}(\pi)]$. By Lemma 2 (single crossing on the up–wedge), there exists a unique boundary $\theta^*(u) \in (\underline{\theta}_0, u)$ such that $\Lambda(\theta^*(u); u) = 0$. Define the pooling mass below the bar by

$$P(u) \ :=\ \int_{\theta^*(u)}^{u} f(\theta)\, d\theta,$$

which is strictly positive because $f > 0$ on the support (Assumption 1) and $\theta^*(u) < u$. The boundary is differentiable in u on the corridor by the implicit–function theorem applied to $\Lambda(\theta; u) = 0$ with $\Lambda_\theta(\theta^*(u); u) > 0$ (again Lemma 2).

Differentiate $P(u) = \int_{\theta^*(u)}^{u} f(\theta)\, d\theta$ using the variable–limits Leibniz rule:

$$P'(u) = f(u) \cdot 1 \ -\ f\big(\theta^*(u)\big) \cdot \theta^{*\prime}(u) = f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u).$$

By GDWBR on the corridor,

$$\theta^{*\prime}(u) \ \geq\ \frac{f(u)}{f(\theta^*(u))}.$$

Substitute this into $P'(u)$ to obtain

$$P'(u) \;\le\; f(u) - f(\theta^*(u)) \cdot \frac{f(u)}{f(\theta^*(u))}$$

with strict inequality on any subinterval where GDWBR is strict or f is not everywhere flat.

Integrate $P'(u) \le 0$ over $u \in [u_{\mathrm{MoM}},\, u_{\mathrm{SM}}(\pi)]$:

$$P\big(u_{\mathrm{SM}}(\pi)\big) - P\big(u_{\mathrm{MoM}}\big) = \int_{u_{\mathrm{MoM}}}^{u_{\mathrm{SM}}(\pi)} P'(u)\, du \;\le\; 0,$$

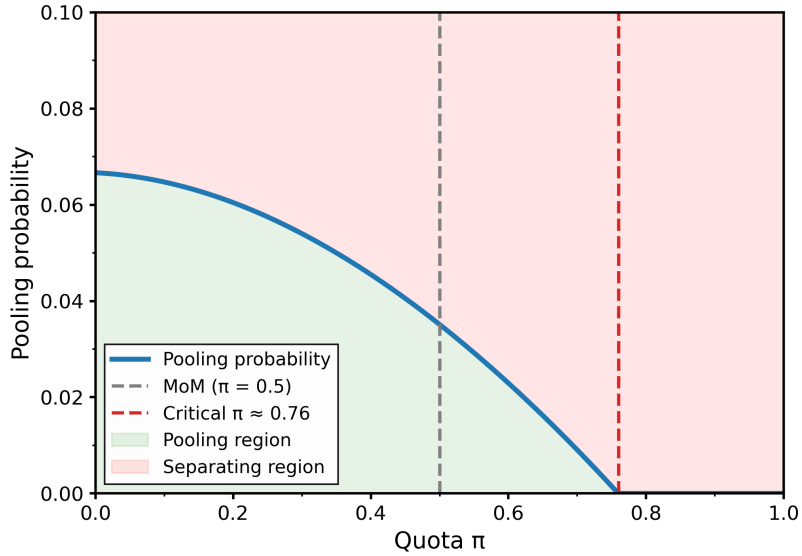
with strict inequality under the nondegeneracy noted in the expression $P'(u) \;\le\; f(u) - f(\theta^*(u)) \cdot \frac{f(u)}{f(\theta^*(u))}$. Rearranging yields $P(u_{\mathrm{SM}}(\pi)) \le P(u_{\mathrm{MoM}})$, as claimed. □

A strict super-majority raises the belief cut-off u above MoM, so types just below the bar must send a stronger persuasive message to pass. Under Assumption 4, minorities' (sluggish/attenuated) updating makes the marginal persuader's best-response boundary $\theta^*(u)$ move at least as fast as the bar in the density-weighted sense (i.e., the behavioral response dominates the purely mechanical "mass at the bar" effect). In the bar–boundary accounting $P'(u) = f(u) - f(\theta^*(u))\, \theta^{*\prime}(u)$, this dominance implies $P'(u) \le 0$: as the rule tightens (higher u), the lower endpoint of the pooling band chases the bar quickly enough that the interval $[\theta^*(u), u]$ shrinks. Along any corridor where partial pooling persists, the stricter SM therefore sustains less pooling than MoM.

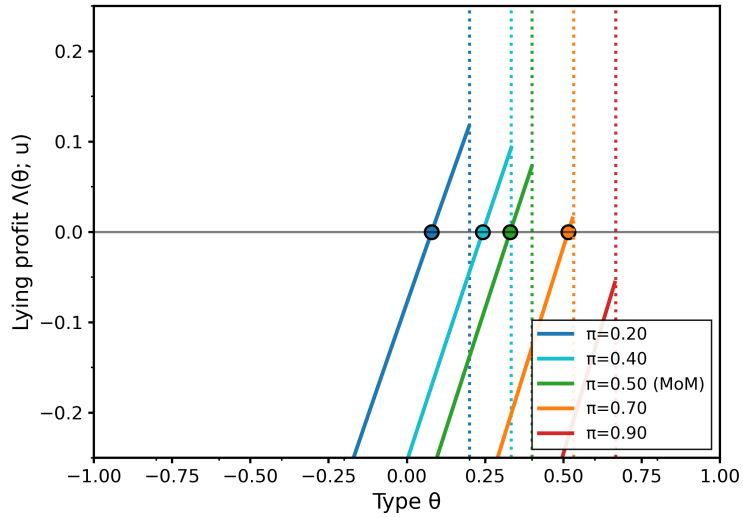
Figure 2 illustrates the GDWBR case: the boundary's faster movement closes the gap $u - \theta^*(u)$, so pooling decreases when tightening from lenient SM to MoM and from MoM to strict SM, until the band vanishes.

Figure 2: Pooling Probabilities and Lying-Profit Curves (Global DWBR)

(a) Panel A: Pooling Probabilities



(b) Panel B: Lying-Profit Curves



Panel A plots the simulated pooling probability $P(u(\pi))$ (solid blue) under a uniform type distribution on $[\theta_0, \theta_1] = [-1, 1]$ with sluggish beliefs $\mu(m) = Km$, where Assumption 4 holds. The controller's cost is $C(m, \theta) = \eta|m - \theta| + \frac{\kappa}{2}(m - \theta)^2$; parameter values are $(\alpha, b_0, b_1, \lambda, K, \eta, \kappa) = (0.50, 0.12, 0.20, 0.10, 0.25, 0.20, 0.12)$. The grey dashed line marks the majority-of-the-minority threshold $\pi = 0.5$, the red dashed line marks the critical quota $\pi \approx 0.76$ at which $P(u(\pi))$ hits zero, and the green (pink) shading indicates the pooling (separating) region. Panel B shows the lying-profit function on the up-wedge, $\Lambda(\theta, u) = (\alpha + \beta)\theta - C(u)$, for $\pi \in \{0.20, 0.40, 0.50, 0.70, 0.90\}$; vertical dotted lines mark the induced belief cut-off $u = u_{\text{rule}}(\pi)$, and open markers indicate the type cut-off θ^* where $\Lambda(\theta^*, u) = 0$.

6 Pooling Regions under Different Voting Regimes: Diversion Dominant

6.1 Single-Crossing

As noted earlier, in the diversion–dominant regime ($B'(\theta) > 1 - \alpha$) the approval test reverses to $\mu(m) \leq u$, and persuasion occurs on the *down–wedge* ($\theta \geq u$). On this side,

$$\Lambda_\theta(\theta; u) = \alpha + B'(\theta) - \left[\eta + \kappa\left(\theta - m^{\text{req}}(u)\right)\right]$$

need not have a fixed sign, so $\Lambda(\cdot; u)$ may fail single crossing and the upper–side cutoff need not be unique; standard single–crossing arguments therefore do not apply. To restore a unique, ordered cutoff in this regime, we impose the following sufficient parameter restriction:

$$\alpha + \sup_{\theta \in [u, \overline{\theta}_0]} B'(\theta) \;\leq\; \eta \qquad \text{(for any } \kappa \geq 0). \tag{P4}$$

Under (P4), $\Lambda_\theta(\theta; u) > 0$ on the down–wedge, reestablishing single crossing and uniqueness of the upper cutoff.

Proposition 10 (PSE for a fixed cut-off in the diversion-dominant (talk-down) regime $B'(\theta) > 1 - \alpha$ with single-crossing)**.** *Fix a voting environment that induces an interior belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0)$ via* (V2)–(V3)*. Assume $B \in C^1$ with $B'(\theta) > 1 - \alpha$ on $[\underline{\theta}_0, \overline{\theta}_0]$ (diversion-dominant), μ satisfies* (M1)*, and signaling costs satisfy* (S2)–(S4)*. Let $m^{\text{req}}(u) := \mu^{-1}(u)$ and define the (minimal talk-down) lying-profit*

$$\Lambda^{\downarrow}(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Upper-side single-crossing ($\text{SC}_\downarrow$)*. Assume that $\theta \mapsto \Lambda^{\downarrow}(\theta; u)$ is strictly single-crossing above the bar: there exists a unique $\theta^{\downarrow *}(u) \in [u, \overline{\theta}_0]$ with*

$$\Lambda^{\downarrow}\big(\theta^{\downarrow *}(u); u\big) = 0, \qquad \Lambda^{\downarrow}(\theta; u) > 0 \text{ for } \theta \in (u, \theta^{\downarrow *}(u)), \qquad \Lambda^{\downarrow}(\theta; u) < 0 \text{ for } \theta > \theta^{\downarrow *}(u).$$

(For instance, Assumption P4—$\alpha + \sup_{\theta \in [u, \overline{\theta}_0]} B'(\theta) \leq \eta$ with any $\kappa \geq 0$—guarantees $\Lambda^{\downarrow}_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta(m^{\text{req}}(u), \theta) < 0$ on the down–wedge and hence strict single crossing.)

Lower-side participation (no-rejection deviation, $\text{CPC}_\downarrow$)*. For all $\theta \leq u$,*

$$\alpha\theta + B(\theta) \;\geq\; -\inf_{m:\ \mu(m) > u} C(m, \theta),$$

i.e. truthful approval weakly dominates any up–wedge deviation that induces rejection. (A sufficient—stronger—form is $\alpha\theta + B(\theta) \geq 0$ for all $\theta \leq u$.)

Then there exists a Perfect–Sequential Equilibrium (PSE) with sender strategy

$$\sigma(\theta) = \begin{cases} \theta, & \theta \le u \quad \textit{(truth; approved)}, \\ m^{\mathrm{req}}(u), & \theta \in (u, \theta^{\downarrow *}(u)) \quad \textit{(cheapest talk-down; approved)}, \\ \theta, & \theta \ge \theta^{\downarrow *}(u) \quad \textit{(truth; rejected)}. \end{cases}$$

Moreover:

(i) ***Uniqueness of allocation.*** *For the same cut-off u, every PSE implements the project if and only if $\theta \le \theta^{\downarrow *}(u)$.*

(ii) ***Pooling vs. separation.*** *The equilibrium is partially pooling on the* upper *side iff $\theta^{\downarrow *}(u) > u$; otherwise it is fully separating above u.*

Proof. In the diversion-dominant regime $B'(\theta) > 1 - \alpha$, the acceptance test flips: define $\Phi(m) := (1 - \alpha)\mu(m) - \beta(m)$. Then Φ is strictly decreasing in $\mu(m)$, so the receivers' rule is

$$\text{approve} \iff \mu(m) \le u,$$

where u is the belief cut-off induced by (V2)–(V3). By (M1), μ is strictly increasing and invertible, hence among approved messages $\{m : \mu(m) \le u\}$ the *least-cost* one for any $\theta > u$ is the maximal m that still passes, i.e.

$$m^{\mathrm{req}}(u) = \mu^{-1}(u).$$

Indeed, for fixed $\theta > u$ the down–wedge gap is $\Delta = m - \theta < 0$; (S3)–(S4) imply C is convex in Δ and strictly increasing in $|\Delta|$, while Δ is minimized over $\{m : \mu(m) \le u\}$ exactly at $m = \mu^{-1}(u)$. Define the lying-profit from minimal talk-down $\Lambda^{\downarrow}(\theta; u) = \alpha\theta + B(\theta) - C(m^{\mathrm{req}}(u), \theta)$. Assumption $\mathrm{SC}_{\downarrow}$ ensures a unique upper-side boundary $\theta^{\downarrow *}(u)$ with the stated sign pattern.

Types at or below the bar. Let $\theta \le u$. Truthful play yields $\mu(\theta) \le u$ and is approved at zero cost by (S2), giving payoff $\alpha\theta + B(\theta)$. Any *approved* deviation has weakly higher signaling cost and leaves the prize unchanged, so it weakly lowers payoff. Any deviation that *induces rejection* must satisfy $\mu(m) > u$ and hence lies on the up–wedge ($\Delta = m - \theta > 0$); by (S3)–(S4) this entails strictly positive cost. By $\mathrm{CPC}_{\downarrow}$, truthful approval weakly dominates such rejecting deviations:

$$\alpha\theta + B(\theta) \ \ge\ -\inf_{\mu(m) > u} C(m, \theta) \ \Rightarrow\ \alpha\theta + B(\theta) \ \ge\ -C(m, \theta)$$

for all rejecting m. Therefore truth is optimal for every $\theta \le u$.

Upper-side persuaders. Fix $\theta \in (u, \theta^{\downarrow *}(u))$. By $\mathrm{SC}_{\downarrow}$, $\Lambda^{\downarrow}(\theta; u) > 0$. Truth would be rejected (since $\mu(\theta) > u$) and yields payoff 0. Sending $m^{\mathrm{req}}(u)$ secures approval at the least cost and yields strictly higher payoff $\Lambda^{\downarrow}(\theta; u) > 0$. Any alternative approved message must satisfy $\mu(m) \le u$ with $m < m^{\mathrm{req}}(u)$ (deeper talk-down), which strictly increases the down–wedge gap and thus signaling cost by (S3)–(S4), so $m^{\mathrm{req}}(u)$ is strictly optimal among approved deviations.

High types. If $\theta > \theta^{\downarrow *}(u)$, then $\Lambda^{\downarrow}(\theta; u) < 0$. Hence even the least-cost approved deviation $m^{\text{req}}(u)$ delivers payoff below truthful rejection (0). Any still-deeper talk-down only raises cost further, so truth with rejection is strictly optimal.

Beliefs and receivers. Given σ from the aforementioned types, on-path messages are either truthful (separating) or the single pooling message $m^{\text{req}}(u)$. Receivers approve iff $\mu(m) \leq u$ by construction. Off path, impose GP-credible beliefs (as in the baseline analysis) so that any unexpected message is interpreted in favor of the types that benefit most from it; no profitable deviation arises given the arguments above.

(i) Uniqueness of allocation for a fixed u. Let $(\hat{\sigma}, \widehat{\text{beliefs}})$ be any PSE under the same cut-off u. Suppose some $\theta > \theta^{\downarrow *}(u)$ is implemented under $(\hat{\sigma}, \widehat{\text{beliefs}})$. Then there exists an approved message $\hat{m}$ with $\mu(\hat{m}) \leq u$ such that $\alpha\theta + B(\theta) - C(\hat{m}, \theta) \geq 0$. But $m^{\text{req}}(u)$ is least-cost among all approved messages, so $\alpha\theta + B(\theta) - C(m^{\text{req}}(u), \theta) \geq 0$, contradicting $\Lambda^{\downarrow}(\theta; u) < 0$. Conversely, suppose some $\theta < \theta^{\downarrow *}(u)$ is not implemented. Then deviating to $m^{\text{req}}(u)$ yields $\Lambda^{\downarrow}(\theta; u) \geq 0$, strictly improving over rejection—contradiction. Hence, for any PSE with this u, the project is implemented iff $\theta \leq \theta^{\downarrow *}(u)$.

(ii) Pooling vs. separation. If $\theta^{\downarrow *}(u) > u$, the interval $(u, \theta^{\downarrow *}(u))$ has positive measure and forms an upper-side pooling region at the common message $m^{\text{req}}(u)$. If $\theta^{\downarrow *}(u) = u$, no type above u finds talk-down profitable and the equilibrium is fully separating above u. □

Proposition 11 (Talk–down GDWBR ordering: Lenient SM $\succ$ MoM $\succ$ Strict SM in pooling mass)**.** *Maintain* (P1)–(P3), (S1)–(S4), (M1), *and* (P4). *Work in the* talk–down *regime (approval iff $\mu(m) \leq u$). For each interior cut–off $u \in (\underline{\theta}_0, \overline{\theta}_0)$ define the minimal approved message $m^{\text{req}}(u) := \mu^{-1}(u)$ and the lying–profit from minimal talk–down*

$$\Lambda^{\downarrow}(\theta; u) \;:=\; \alpha\,\theta + B(\theta) \;-\; C\big(m^{\text{req}}(u), \theta\big).$$

Assume (strict) single–crossing of $\Lambda^{\downarrow}(\cdot; u)$ on the upper side so that for each interior u there exists a unique boundary $\theta^{\downarrow *}(u) \in (u, \overline{\theta}_0)$ with $\Lambda^{\downarrow}(\theta^{\downarrow *}(u); u) = 0$ and $u \mapsto \theta^{\downarrow *}(u)$ is differentiable with $\theta^{\downarrow *\prime}(u) > 0$. Let the rule–induced cut–offs satisfy the monotone selection *ordering*

$$u_{\text{SM}}^{strict} \;<\; u_{\text{MoM}} \;<\; u_{\text{SM}}^{lenient},$$

and assume partial pooling *persists for every interior u on both corridors $[\,u_{\text{SM}}^{strict}, u_{\text{MoM}}\,]$ and $[\,u_{\text{MoM}}, u_{\text{SM}}^{lenient}\,]$ (so $\theta^{\downarrow *}(u) > u$ there). Define the upper pooling mass*

$$Q(u) \;:=\; F\big(\theta^{\downarrow *}(u)\big) - F(u) \;=\; \int_u^{\theta^{\downarrow *}(u)} f(\theta)\, d\theta.$$

If the Global Density–Weighted Boundary Responsiveness *(GDWBR) condition holds on both corridors,*

$$\theta^{\downarrow *\prime}(u) \;\geq\; \frac{f(u)}{f\big(\theta^{\downarrow *}(u)\big)} \qquad \textit{for all interior } u \textit{ on the two corridors,}$$

then the pooling masses are ordered as

$$Q\big(u_{\mathrm{SM}}^{lenient}\big) \;\geq\; Q\big(u_{\mathrm{MoM}}\big) \;\geq\; Q\big(u_{\mathrm{SM}}^{strict}\big).$$

Moreover, each inequality is strict whenever GDWBR is strict on a set of positive measure on the corresponding corridor or the prior density f is not everywhere flat there.

Proof. By (M1) and the talk–down orientation, approval occurs iff $\mu(m) \leq u$ and the least–cost approved message for any $\theta > u$ is the boundary message $m^{\mathrm{req}}(u) = \mu^{-1}(u)$ because the down–wedge cost C is convex in the gap and strictly increases in $|\Delta| = |m - \theta|$ by (S2)–(S4). Under (P4) the upper–side strict single–crossing yields a unique boundary $\theta^{\downarrow *}(u) \in (u, \overline{\theta}_0)$ with $\Lambda^{\downarrow}(\theta^{\downarrow *}(u); u) = 0$ and $\theta^{\downarrow *\prime}(u) > 0$ via the implicit–function theorem. On the corridors, partial pooling implies $\theta^{\downarrow *}(u) > u$ and hence

$$Q(u) \;=\; \int_{u}^{\theta^{\downarrow *}(u)} f(\theta)\, d\theta \;>\; 0.$$

Differentiate $Q(u)$ using the variable–limits Leibniz rule:

$$Q'(u) \;=\; f\big(\theta^{\downarrow *}(u)\big)\, \theta^{\downarrow *\prime}(u) \;-\; f(u).$$

GDWBR gives

$$\theta^{\downarrow *\prime}(u) \;\geq\; \frac{f(u)}{f\big(\theta^{\downarrow *}(u)\big)}.$$

Substitute into $Q'(u)$ to obtain $Q'(u) \geq 0$, with strict inequality whenever GDWBR is strict or f is not everywhere flat on a subset of the corridor.

First, along $[\, u_{\mathrm{MoM}}, u_{\mathrm{SM}}^{\mathrm{lenient}} \,]$ we have

$$Q\big(u_{\mathrm{SM}}^{\mathrm{lenient}}\big) - Q\big(u_{\mathrm{MoM}}\big) \;=\; \int_{u_{\mathrm{MoM}}}^{u_{\mathrm{SM}}^{\mathrm{lenient}}} Q'(u)\, du \;\geq\; 0.$$

Second, along $[\, u_{\mathrm{SM}}^{\mathrm{strict}}, u_{\mathrm{MoM}} \,]$ we have

$$Q\big(u_{\mathrm{MoM}}\big) - Q\big(u_{\mathrm{SM}}^{\mathrm{strict}}\big) \;=\; \int_{u_{\mathrm{SM}}^{\mathrm{strict}}}^{u_{\mathrm{MoM}}} Q'(u)\, du \;\geq\; 0.$$

Strictness in either difference follows from the nondegeneracy noted in both $Q\big(u_{\mathrm{SM}}^{\mathrm{lenient}}\big) - Q\big(u_{\mathrm{MoM}}\big) = \int_{u_{\mathrm{MoM}}}^{u_{\mathrm{SM}}^{\mathrm{lenient}}} Q'(u)\, du \geq 0$ and $Q\big(u_{\mathrm{MoM}}\big) - Q\big(u_{\mathrm{SM}}^{\mathrm{strict}}\big) = \int_{u_{\mathrm{SM}}^{\mathrm{strict}}}^{u_{\mathrm{MoM}}} Q'(u)\, du \geq 0$. Combining the two inequalities yields the asserted chain $Q(u_{\mathrm{SM}}^{\mathrm{lenient}}) \geq Q(u_{\mathrm{MoM}}) \geq Q(u_{\mathrm{SM}}^{\mathrm{strict}})$. □

In the talk–down regime, approval requires $\mu(m) \leq u$, so a more *lenient* rule raises the belief cut–off u (expands the set approved at truth), while a *strict* rule lowers it. The upper pooling

mass is $Q(u) = \int_u^{\theta^{\downarrow *}(u)} f(\theta)\, d\theta$, where $\theta^{\downarrow *}(u)$ is the marginal type who is just indifferent between truthful rejection and minimally talking down to $m_{\text{req}}(u) = \mu^{-1}(u)$. Like what was discussed earlier in Section 5.1.3.1, raising u has two countervailing effects on $Q(u)$: mechanically it subtracts mass at the lower endpoint (the "bar effect"), but behaviorally it shifts the indifference boundary upward (the "boundary effect"). Under GDWBR, the boundary responds *at least as fast* in density–weighted terms as the bar moves, so the boundary effect dominates and $Q'(u) \geq 0$. Hence a lenient SM (higher u) yields *more* pooling than MoM, while a strict SM (lower u) yields *less* pooling than MoM. Thus, where single-crossing holds, we have very similar results to the up-talk regime cases discussed earlier in the article.

6.2 No Single-Crossing

In the absence of single crossing, the upper–side cutoff need not be unique, and the induced approval set need not vary monotonically with type θ; consequently, the rule–induced bar u need not be isotone in θ. The next result makes this precise.

Proposition 12 (PSE at a fixed cut–off without single–crossing (talk–down))**.** *Fix a voting environment that induces an interior belief cut–off $u \in (\underline{\theta}_0, \overline{\theta}_0)$ via* (V2)–(V3)*. Assume $B \in C^1$ with $B'(\theta) > 1 - \alpha$ on $[\underline{\theta}_0, \overline{\theta}_0]$, μ satisfies* (M1)*, and costs satisfy* (S2)–(S4)*. Let $m^{\text{req}}(u) := \mu^{-1}(u)$ and define the* lying-profit from minimal talk–down

$$\Lambda^{\downarrow}(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Writing the upper wedge by $\Theta^{\uparrow}(u) := (u, \overline{\theta}_0]$, define the approved-by-talk-down set

$$\mathcal{A}(u) \;:=\; \big\{\theta \in \Theta^{\uparrow}(u) \;:\; \Lambda^{\downarrow}(\theta; u) \geq 0\big\}.$$

(Without single–crossing, $\mathcal{A}(u)$ may be empty, an interval, or a finite union of disjoint intervals.)

Claim. *There exists a Perfect–Sequential Equilibrium (PSE) with sender strategy*

$$\sigma_u(\theta) = \begin{cases} \theta, & \theta \leq u \quad \textit{(truth; approved)}, \\ m^{\text{req}}(u), & \theta \in \mathcal{A}(u) \quad \textit{(least-cost talk–down; approved)}, \\ \theta, & \theta \in \Theta^{\uparrow}(u) \setminus \mathcal{A}(u) \quad \textit{(truth; rejected)}. \end{cases}$$

Receivers approve iff $\mu(m) \leq u$. Hence the implemented set *at u is*

$$\mathcal{I}(u) \;=\; (-\infty,\, u] \;\cup\; \mathcal{A}(u).$$

Moreover:

(i) ***Maximal pooling at*** $m^{\text{req}}(u)$**.** *In any PSE in which types above u ever send $m^{\text{req}}(u)$ on path,*

the approved upper-side set must equal $\mathcal{A}(u)$. In particular, if some $\tilde{\theta} \in \mathcal{A}(u)$ is approved via $m^{\text{req}}(u)$, then every $\theta \in \mathcal{A}(u)$ is (profitably) drawn into the same pool.

(ii) ***Minimal pooling (existence).*** *There also exists a PSE with* no *upper-side pooling (i.e., $\sigma_u(\theta) = \theta$ and rejection for all $\theta > u$), supported by GP-credible off-path beliefs at any down-deviation $m < m^{\text{req}}(u)$ that assign posterior mean strictly above u, so the deviation is not approved.*

Proof. When $B'(\theta) > 1 - \alpha$, the approval statistic $(1 - \alpha)\mu - \beta$ is strictly decreasing in μ, so the test is "approve iff $\mu(m) \leq u$"; the smallest-cost approved message is $m^{\text{req}}(u) := \mu^{-1}(u)$ by (M1). On the down wedge ($m < \theta$), (S4) gives $C_m(m, \theta) = -\eta + \kappa(m - \theta) < 0$: any *further* talk–down (lower m) strictly raises cost; hence $m^{\text{req}}(u)$ is the unique least-cost way to pass the bar for any $\theta > u$.

Types at or below the bar. If $\theta \leq u$, then truth $m = \theta$ is approved at zero cost; any lie adds cost and cannot help. Thus $\sigma_u(\theta) = \theta$ is optimal and approved.

Upper wedge: profitable vs. unprofitable talk–down. Fix $\theta > u$. If $\theta \in \mathcal{A}(u)$, then $\Lambda^{\downarrow}(\theta; u) \geq 0$, so deviating to the least-cost approved message $m^{\text{req}}(u)$ weakly dominates truth (which is rejected with payoff 0). Any other approved message $m < m^{\text{req}}(u)$ is strictly more costly by the wedge monotonicity above and is therefore weakly dominated by $m^{\text{req}}(u)$. If $\theta \notin \mathcal{A}(u)$, then $\Lambda^{\downarrow}(\theta; u) < 0$; any larger lie $m < m^{\text{req}}(u)$ is even more costly, so every approved deviation yields a payoff < 0, strictly dominated by truthful rejection with payoff 0. Hence $\sigma_u(\theta) = \theta$.

Beliefs and receivers. On path, messages are either truth $m = \theta$ or $m^{\text{req}}(u)$. Receivers approve exactly those m with $\mu(m) \leq u$ by construction: truth below u and the pooled $m^{\text{req}}(u)$.

For the *maximal-pooling* claim (i), suppose some upper-side type approves via $m^{\text{req}}(u)$ on path. Then, for any $\theta \in \mathcal{A}(u)$, deviating to that same on-path message yields payoff $\Lambda^{\downarrow}(\theta; u) \geq 0$ and approval; hence any PSE that approves some member of $\mathcal{A}(u)$ via $m^{\text{req}}(u)$ must approve all of $\mathcal{A}(u)$.

For the *minimal-pooling* existence (ii), consider a profile with $\sigma_u(\theta) = \theta$ for all θ and the receiver approving iff $\mu(m) \leq u$. At any down-deviation $m < m^{\text{req}}(u)$, adopt GP-credible beliefs (Grossman–Perry) that concentrate on the set of types that benefit most from such a deviation. These types must lie on the upper wedge ($\theta > u$) (since $\theta \leq u$ are already approved at zero cost and would strictly lose by paying for a lie); thus the posterior mean at such m exceeds u, and the deviation is rejected.[9] All remaining off-path deviations are weakly dominated by truth. Thus no type above u wishes to deviate and the no-pooling PSE exists. □

Without single–crossing, the talk–down approval set $\mathcal{A}(u)$ need not be a single interval: $\Lambda(\cdot; u)$ can admit multiple roots, generating alternating pockets of pooling and separation. Even so, the *least–cost* property of the minimal persuasive message $m_{\text{req}}(u)$ imposes a stark selection at any

[9] Because $\mu(\cdot)$ is strictly increasing (M1) and the credible-beliefs set is a subset of $(u, \overline{\theta}_0]$, the induced $\mu(m)$ can be chosen $> u$ while keeping μ strictly increasing globally; small local adjustments suffice around the off-path point.

such bar: either all types in $\mathcal{A}(u)$ coordinate on pooling at $m_{\text{req}}(u)$ (maximal pooling), or none do (minimal pooling sustained by credible off–path beliefs). Which equilibrium is selected depends on beliefs, not on first–order payoffs, so comparative statics in u are generally ambiguous absent additional structure. In the example that follows, we illustrate how fixed primitives (priors and parameters) can produce alternating pockets of pooling across the type space θ.

Example 1 (Talk–down regime with (S1) costs: quadratic in $x = \theta - u$ on the down–wedge)**.** ***Primitives*** *Prior: uniform on* $[\underline{\theta}_0, \overline{\theta}_0] = [-1, 1]$. *Beliefs:* $\mu(m) = m$. *Private benefits:* $B(\theta) = b_0 + b_1\theta$ *with*

$$\alpha = 0.3, \quad b_1 = 0.8\ (> 1 - \alpha = 0.7), \quad b_0 = -0.02.$$

Signalling cost of type (S1):

$$C(m, \theta) = \eta\,|m - \theta| + \frac{\kappa}{2}(m - \theta)^2, \qquad \eta = 1,\ \kappa = 2.$$

Because $b_1 > 1-\alpha$, *we are in the talk–down case: approval iff* $\mu(m) \le u$, *hence the minimal approved message is* $m^{\text{req}}(u) = u$.

Rule cut–offs and CPC *With acceptance condition*

$$(1 - \alpha)\,u - \beta(u) \;=\; \underline{\kappa}_{\text{rule}} \quad \Longrightarrow \quad u = \frac{\underline{\kappa}_{\text{rule}} + b_0}{1 - \alpha - b_1} = \frac{\underline{\kappa}_{\text{rule}} - 0.02}{-0.1}.$$

Take:

$$\underline{\kappa}_{\text{MoM}} = 0 \;\Rightarrow\; u_{\text{MoM}} = 0.2, \qquad \underline{\kappa}_{\text{SM}}^{strict} = +0.01 \;\Rightarrow\; u_{\text{str}} = 0.1, \qquad \underline{\kappa}_{\text{SM}}^{lenient} = -0.01 \;\Rightarrow\; u_{\text{len}} = 0.3.$$

Controller–participation: $(\alpha + b_1)\,u + b_0 = 1.1\,u - 0.02 > 0$ *for each of* $u = 0.1, 0.2, 0.3$, *so CPC holds.*

Lying–profit on the down–wedge *For* $\theta \ge u$, *set* $x := \theta - u \ge 0$. *Then*

$$\Lambda(\theta; u) = \alpha\,\theta + B(\theta) - C\big(u, \theta\big) = (\alpha + b_1)\,\theta + b_0 - \eta\,(\theta - u) - \tfrac{\kappa}{2}\,(\theta - u)^2.$$

In terms of x *it becomes*

$$\Lambda(u + x; u) = A\,u + b_0 + (A - \eta)\,x - \frac{\kappa}{2}\,x^2, \qquad A := \alpha + b_1 = 1.1.$$

Hence the nonnegative root $x_+(u)$ *is*

$$x_+(u) = \frac{(A - \eta) + \sqrt{(A - \eta)^2 + 2\,\kappa\,(A\,u + b_0)}}{\kappa}, \qquad \theta^{\downarrow *}(u) \;=\; u + x_+(u).$$

Numerical evaluation (support $[-1, 1]$***)***

Strict SM ($u = 0.1$): With $A = 1.1$,

$$x_+(0.1) = \frac{(1.1-1) + \sqrt{(0.1)^2 + 2 \cdot 2 \cdot (1.1 \cdot 0.1 - 0.02)}}{2} \approx 0.354,$$

so $\theta^{\downarrow *}(0.1) \approx 0.1 + 0.354 = 0.454$. Upper pooling interval: $(0.1,\ 0.454)$ of length 0.354, pooling mass $Q_{\text{str}} \approx 0.354/2 = 0.177$.

MoM ($u = 0.2$):

$$x_+(0.2) = \frac{(1.1-1) + \sqrt{0.1^2 + 2 \cdot 2 \cdot (1.1 \cdot 0.2 - 0.02)}}{2} \approx 0.500,$$

so $\theta^{\downarrow *}(0.2) \approx 0.700$. Pooling interval: $(0.2,\ 0.700)$ of length 0.500, pooling mass $Q_{\text{MoM}} = 0.250$.

Lenient SM ($u = 0.3$):

$$x_+(0.3) \approx 0.609, \qquad \theta^{\downarrow *}(0.3) \approx 0.909.$$

Pooling interval: $(0.3,\ 0.909)$ of length 0.609, pooling mass $Q_{\text{len}} \approx 0.609/2 = 0.305$.

Pooling Mass Comparisons

$$Q_{\text{len}} \approx 0.305 \ > \ Q_{\text{MoM}} = 0.250 \ > \ Q_{\text{str}} \approx 0.177.$$

In each case, $\Lambda(\theta; u)$ has the unique root above u given by $\theta^{\downarrow *}(u) = u + x_+(u)$, and CPC holds.

7 Conclusion

This paper has developed a new informational framework for understanding how voting thresholds shape the communication strategies of controlling shareholders in conflicted transactions. By modelling a controller who privately observes project quality and can cost-effectively distort disclosures, we show that voting thresholds not only determine whether a proposal succeeds but also whether the underlying types pool or separate in equilibrium. Under responsive beliefs and standard priors, a lenient supermajority yields the smallest pooling region, a moderate majority generates an intermediate region, and a stringent supermajority the largest. However, when beliefs are sluggish—due to noisy signaling, rational inattention, or endogenous non-participation—this ordering reverses: pooling becomes largest under the lenient rule and smallest under the stringent rule. In a diversion-dominant regime, where private benefits to the controller dominate minorities' monitoring power, multiple equilibria arise and the rule ranking becomes ambiguous.

At root is a bar–boundary trade-off: raising the bar mechanically lifts the mass of types above it, but also shifts the indifferent boundary of pooling behavior. The appropriate voting rule should therefore be matched to the information environment—priors, belief responsiveness, and signaling costs—since these factors determine which side of the trade-off dominates. This analysis is or-

thogonal to the standard Type-I/Type-II error trade-off in conflicted-transaction review; those error costs should be considered alongside (and may reinforce) the environment-contingent choice of rule. One policy takeaway is that we should use controller-exclusion (MoM) as the baseline procedural safeguard and set the quota (lenient versus strict supermajority) to minimize pooling given the observed information environment.

Several avenues for future research beckon. First, allowing ownership to be endogenous would shed light on how controlling shareholders anticipate and exploit voting rules by adjusting stake size. Second, modelling strategic minority voting or active information acquisition could enrich the communication game and reveal how minority coordination affects pooling and separation. Third, extending the model to multidimensional projects or investment levels would show how thresholds influence not only approval, but project design and exaggeration. Fourth, incorporating reputational or intermediary mechanisms (such as proxy advisors or monitoring analysts) would deepen our understanding of how disclosure quality emerges. Finally, the empirical predictions—namely that the size of pooling regions varies systematically with belief responsiveness and threshold rigidity—await testing using cross-jurisdictional or quasi-natural variation in voting rules.

As controlling shareholders remain central in global corporate governance, this work highlights that voting rules do more than aggregate votes—they fundamentally influence what information gets produced, transmitted, and acted upon in the boardroom. As controlling shareholders remain central in global corporate governance, voting rules do more than aggregate votes—they reshape the production and transmission of information in the boardroom.

A Appendix

A.1 Additional Proofs

Corollary 1 (Degenerate voting rules)**.** *Let $u = \underline{\theta}$ be the rule–specific belief cut-off.*

(i) ***Trivially easy rule*** $\left[u < \underline{\theta}_0\right]$*. Every posterior belief satisfies $\mu(m) \geq u$, hence minorities always vote "Yes". For the controller any message in the approval region is accepted; because $C(m, \theta) \geq 0$ with equality at $m = \theta$, truthful reporting maximises payoff but* any *message weakly below the truth is also a best reply. Consequently an uncountable continuum of Perfect–Sequential Equilibria exists, ranging from complete pooling $\sigma(\theta) \equiv m_L$ (with any $m_L \leq u$) to full separation $\sigma(\theta) = \theta$. The pooling probability can take any value in* $[0, 1]$.

(ii) ***Trivially tough rule*** $\left[u > \overline{\theta}_0\right]$*. No posterior belief can reach the hurdle, so minorities always vote "No". The controller's unique best reply is to choose the cheapest message, i.e. $\sigma(\theta) = \theta$ (because $C(m, \theta)$ is minimised at $m = \theta$). The unique PSE is therefore fully separating with universal rejection.*

Lemma 5 (No single-message pooling)**.** *Fix an interior belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0)$, and maintain Assumptions* P1–P3, S1–S4, M1*. Off the equilibrium path, beliefs satisfy the Grossman–Perry credible-*

beliefs refinement. Then there is no *Perfect–Sequential Equilibrium (PSE) in which every* $\theta \in (\underline{\theta}_0, \overline{\theta}_0)$ *sends the same message* $m^\flat$*, regardless of whether minorities would approve or reject* $m^\flat$*.*

Proof. Let $m^{\text{req}} := \mu^{-1}(u)$ denote the minimal persuasive message (V3). Two cases:

(a) Common message is rejected: $\mu(m^\flat) < u$**.** Then minorities always vote "No" and the controller's equilibrium payoff is $U^{\text{eq}}(\theta) = -C(m^\flat, \theta)$. Pick a high type $\bar{\theta}$ close enough to $\overline{\theta}_0$ such that

$$\alpha\bar{\theta} + B(\bar{\theta}) \;>\; C\big(m^{\text{req}}, \bar{\theta}\big).$$

This is feasible because B is strictly increasing (P3) and, by the type-gradient (S4), the signaling cost weakly *decreases* in θ at any fixed message. If type $\bar{\theta}$ deviates to the approved message m^{req} she obtains

$$U^{\text{dev}} = \alpha\bar{\theta} + B(\bar{\theta}) - C\big(m^{\text{req}}, \bar{\theta}\big) \;>\; 0 \;\geq\; U^{\text{eq}}(\bar{\theta}),$$

contradicting sequential rationality.

(b) Common message is approved: $\mu(m^\flat) \geq u$**.** We split by the location of $m^\flat$.

(b1) $m^\flat < \overline{\theta}_0$ *(some types under-report).* Choose any type $\theta^+ \in (m^\flat, \overline{\theta}_0]$ and fix a small $\varepsilon > 0$; consider the deviation $m^+ := m^\flat + \varepsilon$. Because the deviator is under-reporting at $m^\flat$, (S3) implies $C_m(m, \theta^+) = -\eta(\theta^+) < 0$ for all $m < \theta^+$, hence

$$C(m^+, \theta^+) < C(m^\flat, \theta^+).$$

Under the Grossman–Perry refinement, upon observing the *higher* off-path message m^+, minorities assign probability 1 to the set of types who benefit most from this deviation. Only *high* types with $\theta \geq m^\flat$ strictly benefit from moving up, so the posterior at m^+ is supported on $\{\theta \geq m^\flat\}$. Since the minority's approval statistic

$$g(\theta) := (1-\alpha)\theta - B(\theta)$$

is strictly increasing in θ (P3), the posterior at m^+ first-order stochastically *dominates* the pooling posterior at $m^\flat$, and thus $\mathbb{E}[g(\theta) \mid m^+] \geq \mathbb{E}[g(\theta) \mid m^\flat]$. Because the project is approved at $m^\flat$, it remains approved at m^+ as well. The deviator's benefit is unchanged (approval), while her cost is strictly lower, so $U^{\text{dev}}(\theta^+) > U^{\text{eq}}(\theta^+)$—a contradiction.

(b2) $m^\flat \geq \overline{\theta}_0$ *(all types over-report).* Evaluate the pooling payoff at the bottom type. By normalization $B(\underline{\theta}_0) = 0$ (P3) and since $m^\flat > \underline{\theta}_0$, (S2)–(S3) give $C(m^\flat, \underline{\theta}_0) > 0$. Because $\underline{\theta}_0 < 0$,

$$U^{\text{eq}}(\underline{\theta}_0) = \alpha\,\underline{\theta}_0 + B(\underline{\theta}_0) - C(m^\flat, \underline{\theta}_0) \;<\; 0.$$

This type strictly prefers truthful rejection (payoff 0), so a single-message approved pooling profile cannot be a PSE.

Combining (a), (b1), and (b2), no PSE can feature a single message used by all types when the cut-off u is interior. □

A.2 Credible-Beliefs Refinement (Grossman–Perry 1986)

A.2.1 Definition

Consider any signalling game with Perfect Bayesian Equilibria (PBE). Let m_{off} be a *message that is not sent with positive probability in the candidate equilibrium.* A PBE satisfies the credible-beliefs refinement if the receiver's belief $\mu(\cdot \mid m_{\text{off}})$ assigns probability 1 to the set

$$T^{\text{best}}(m_{\text{off}}) := \Big\{ \theta \in \Theta : \; U_S\big(m_{\text{off}}, \theta\big) \;\geq\; U_S\big(m', \theta\big) \;\; \forall m' \text{ in the equilibrium support} \Big\},$$

i.e. the set of sender types that would (weakly) benefit *most* from deviating to m_{off}. If T^{best} is empty, beliefs are unrestricted.

Intuitively, the receiver should deem the unexpected signal most likely to come from the type with the strongest incentive to deviate, which is usually the *worst* type from the receiver's standpoint. This pessimism discourages profitable deviations and helps select a unique equilibrium.

A.2.2 Illustration in Our Partial-Pooling Equilibrium

Equilibrium recap. In the pooling band where $\theta < \theta^*$, all types send the common message m_L and minority shareholders reject the proposal. In the separating band where $\theta \geq \theta^*$, each type reports truthfully; the minorities approve precisely when $(1-\alpha)\theta - B(\theta) \geq \underline{\kappa}^{(\lambda)}$.

Off-path deviations. Consider a sender who instead transmits a higher message m_H that is not prescribed by the equilibrium. High-quality types ($\theta \geq \theta^*$) already secure approval without cost, so switching to m_H cannot improve their payoff. Low-quality types ($\theta < \theta^*$) are currently rejected and would gain $\alpha\theta + B(\theta) - C(m_H, \theta)$ only if the deviation were accepted. Hence the set of types with the strongest incentive to deviate, $T^{\text{best}}(m_H)$, contains exactly these low types.

Credible beliefs assign probability one to $\theta \in T^{\text{best}}(m_H)$ upon observing m_H. Minority shareholders therefore adopt the most pessimistic posterior, and the inequality $(1-\alpha)\mu(m_H) - \beta(m_H) \geq \underline{\kappa}^{(\lambda)}$ fails. They vote "No." The deviating sender then pays the full signalling cost $C(m_H, \theta)$ without gaining private benefits, making deviation strictly unprofitable.

Because every off-equilibrium message is treated in this way, no type wishes to deviate, and the partial-pooling profile remains the unique perfect sequential equilibrium under the given voting rule. Credible beliefs thus eliminate the need for arbitrary off-path assumptions: outsiders always interpret surprise signals in the manner most damaging to the sender, thereby ruling out equilibria that depend on optimistic beliefs and resolving multiplicity.

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